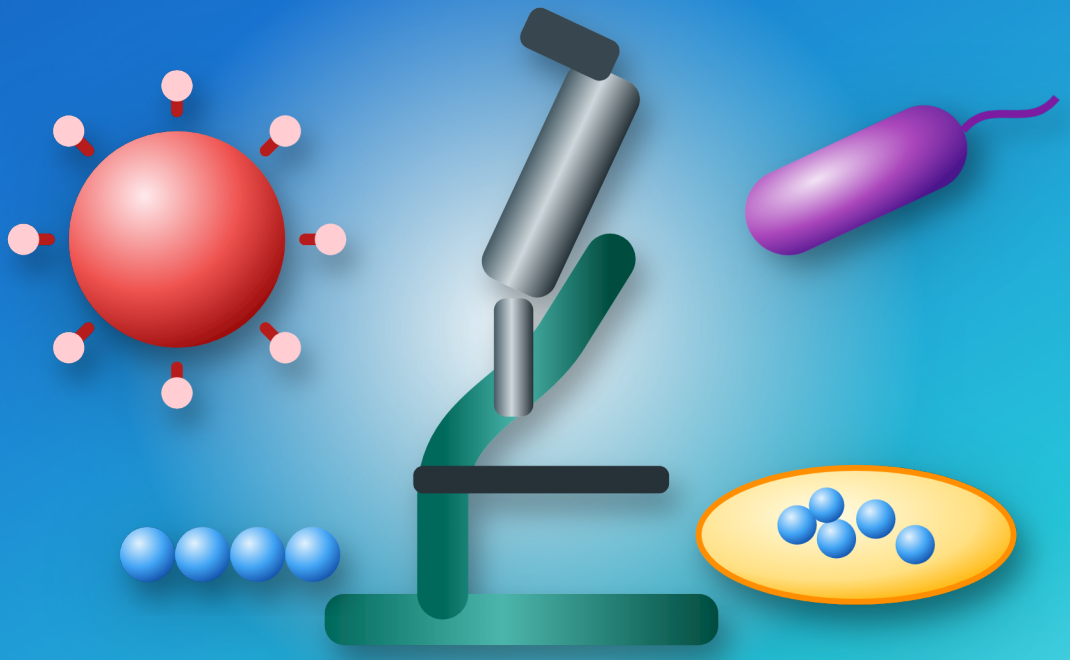
GSSSB · STAFF NURSE EXAM

Microbiology for Nurses

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



10 Chapters | 3D Coloured Diagrams | Bacteria · Viruses · Fungi · Parasites | Sterilisation & infection control | Specimen collection · BMW | Exam Points (PYQ style)

Hinglish explanation + English exam terms

Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **Introduction, History & Microscopy**
Definition & branches, history – Leeuwenhoek, Pasteur, Koch, Lister, Jenner, Fleming; Koch's postulates, classification of microbes, prokaryotes vs eukaryotes, microscope types, normal flora, importance in nursing

2 **Bacterial Morphology, Physiology & Laboratory Methods**
Shapes & arrangement, bacterial cell structure – cell wall, capsule, flagella, pili, spores, plasmids; staining methods; growth requirements, growth curve, culture media, culture methods, bacterial toxins, bacterial genetics & drug resistance

3 **Sterilisation & Disinfection**
Definitions, physical methods (heat, radiation, filtration), autoclave, hot air oven, chemical disinfectants, Spaulding classification, sterilisation indicators, CSSD, care of instruments, fumigation, spill management

4 **Infection, Hospital-Acquired Infections & Infection Control**
Types & sources of infection, chain of infection, modes of transmission, stages of infectious disease, HAI/ HCAI (CAUTI, CLABSI, VAP, SSI), standard & transmission-based precautions, hand hygiene (WHO 5 moments, 7 steps), PPE donning/ doffing, needle-stick injury & PEP, biomedical waste management rules 2016, HICC

5 **Systematic Bacteriology – Important Pathogenic Bacteria**
Gram-positive cocci (Staph, Strep, Pneumococcus, Enterococcus), Gram-negative cocci (Neisseria), Gram-positive bacilli (diphtheria, anthrax, Clostridium), Gram-negative bacilli (E. coli, Klebsiella, Salmonella, Shigella, Vibrio, Pseudomonas, H. influenzae, pertussis, H. pylori), Mycobacteria (TB, leprosy), spirochaetes, Rickettsia, Chlamydia, Mycoplasma

6 **Virology**
Viral structure, classification, replication, cultivation, inclusion bodies, interferon; DNA viruses (herpes, HPV, HBV, pox, adeno); RNA viruses (polio, rotavirus, hepatitis A/C/D/E, measles, mumps, rubella, influenza, RSV, rabies, dengue, chikungunya, JE, HIV, SARS-CoV-2, Nipah); antiviral basics

7 **Mycology & Parasitology**
Fungi – structure, classification, superficial, subcutaneous, systemic & opportunistic mycoses, lab diagnosis; Protozoa – Entamoeba, Giardia, Plasmodium (malaria), Leishmania, Toxoplasma, Trichomonas, Cryptosporidium; Helminths – roundworm, hookworm, pinworm, whipworm, Strongyloides, filaria, Guinea worm, tapeworms, hydatid, flukes; vectors

8 **Immunity, Serology & Vaccines**
Innate & acquired immunity, antigen & antibody, complement, hypersensitivity, antigen-antibody reactions (agglutination, precipitation, ELISA, rapid tests, PCR), important serological tests, types of vaccines, cold chain, UIP schedule, AEFI, immunoglobulins & antisera

9 **Specimen Collection, Transport & Laboratory Diagnosis**
General principles, labelling & requisition, blood culture, urine (midstream, catheter, 24-hr), stool, sputum, throat/ nasal swabs, pus & wound swabs, CSF, high vaginal swab, specimens for virology, transport & storage, rejection criteria, lab safety & biosafety levels, nurse's role

10 **Quick Revision – Master Tables & 150 One-Liners**
Disease–organism table, culture media table, stains, incubation periods, diseases & vectors, important days, 150 rapid-fire one-liners and exam strategy

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Introduction, History & Microscopy

1.1 Definition & Branches

D Definitions

- **Microbiology** – study of **microorganisms (microbes)** too small to be seen by naked eye (< 0.1 mm) – **bacteria, viruses, fungi, protozoa, helminths (parasitology), algae**.
- **Pathogen** – microbe capable of causing disease; **opportunistic pathogen** – causes disease only when host immunity is low (Candida, Pneumocystis); **commensal** – lives on host without harm (normal flora).
- **Virulence** – degree of pathogenicity; **infectivity**, **invasiveness**, **toxigenicity**.
- **Branches** – bacteriology, virology, mycology (fungi), parasitology (protozoa & helminths), immunology, medical/ clinical microbiology.

1.2 History – Important Contributors

Scientist	Contribution
Antonie van Leeuwenhoek	**Father of microbiology**; first observed microbes ("**animalcules**", 1676) with **simple microscope** – **father of bacteriology/ protozoology** too
Louis Pasteur	**Father of modern microbiology**; **disproved spontaneous generation (swan-neck flask)**; **germ theory of disease**; **pasteurisation**; **fermentation**; vaccines – **rabies (1885)**, **anthrax**, chicken cholera; his associate **Chamberland** developed the **autoclave**; term "**vaccine**" (honouring Jenner)
Robert Koch	**Father of bacteriology** (along with Leeuwenhoek in some texts); discovered **Bacillus anthracis (1876), Mycobacterium tuberculosis (1882 – 24 March – World TB Day), Vibrio cholerae (1883)**; **Koch's postulates**; **solid media (agar – suggested by Fanny Hesse)**, staining methods; **Koch phenomenon**
Joseph Lister	**Father of antiseptic surgery** – used **carbolic acid (phenol)** (1867)
Edward Jenner	**Father of immunology/ vaccination** – **smallpox vaccine (cowpox, 1796)**
Alexander Fleming	**Penicillin (1928)** from Penicillium notatum; **lysozyme**
Paul Ehrlich	**Father of chemotherapy**; "**magic bullet**" – **Salvarsan (arsenical) for syphilis**; acid-fast staining (modified)
Ignaz Semmelweis	**Hand washing (chlorinated lime) – reduced puerperal sepsis** – "**saviour of mothers**"
Florence Nightingale	Hygiene & sanitation in hospitals (Crimean war)
Hans Christian Gram	**Gram stain (1884)**
Ziehl & Neelsen	**Acid-fast stain**

Scientist	Contribution
Dmitri Ivanovsky / Beijerinck	**Discovered viruses** (tobacco mosaic virus, 1892); Beijerinck – "contagium vivum fluidum"
Elie Metchnikoff	**Phagocytosis** (cellular immunity)
Emil von Behring	**Antitoxin (diphtheria)** – first Nobel Prize in Medicine (1901)
Selman Waksman	**Streptomycin** (TB); coined "**antibiotic**"
Jonas Salk / Albert Sabin	**Injectable killed (IPV, 1955) / oral live polio vaccine (OPV)**
Ronald Ross	**Malaria transmitted by mosquito (Secunderabad, 1897)** – Nobel 1902
Laveran	Discovered **malarial parasite** in blood
Kary Mullis	**PCR**
Robert Hooke	First described **cells** & fungi (1665)
Carl Woese	Three-domain classification (rRNA)

Koch's postulates

1. The organism must be present in **every case** of the disease (absent in healthy).
2. It must be **isolated in pure culture**.
3. Pure culture **inoculated into a healthy susceptible animal** must produce the **same disease**.
4. The organism must be **re-isolated** from the experimental animal.
   - **Exceptions** – organisms that **can't be cultured (M. leprae, T. pallidum)**, carriers, viruses; **molecular Koch's postulates (Falkow)**.

1.3 Classification of Microorganisms

Feature	Prokaryotes (bacteria)	Eukaryotes (fungi, protozoa, helminths, human cells)
Nucleus	**No nuclear membrane** (nucleoid), single circular chromosome	**True nucleus with membrane**, multiple chromosomes
Organelles	**No mitochondria, ER, Golgi**	Present
Ribosomes	**70S (30S + 50S)** – antibiotic target	**80S (40S + 60S)**
Cell wall	**Peptidoglycan** (except Mycoplasma – no wall)	Fungi – **chitin**; animal cells – none
Sterols in membrane	Absent (except Mycoplasma)	Present (fungi – **ergosterol**, target of azoles & amphotericin)
Division	**Binary fission**	Mitosis/ meiosis
Size	0.5–5 µm	> 5 µm

- **Viruses** – **acellular**, **obligate intracellular parasites**, contain **either DNA or RNA (never both)**, no metabolism/ ribosomes, **not affected by antibiotics**; **prions** – infectious proteins (no nucleic acid – CJD, kuru, mad cow) – **most resistant to sterilisation**; **viroids** – naked RNA (plants).
- **Chlamydia & Rickettsia** – **obligate intracellular bacteria**; **Mycoplasma** – **smallest free-living bacteria, no cell wall** (penicillin-resistant).

- **Whittaker's 5 kingdoms** – Monera (bacteria), Protista, Fungi, Plantae, Animalia; **3 domains** – Bacteria, Archaea, Eukarya.
- **Binomial nomenclature (Linnaeus)** – genus (capital) + species (small), italics – e.g. **Staphylococcus aureus**.

1.4 Microscopy

Microscope	Principle / use
Light (bright-field/ compound) microscope	Most common; stained smears; **magnification up to ~1000× (oil immersion – 100× objective × 10× eyepiece)**; **resolving power ~0.2 µm**; **cedar wood oil/ immersion oil** ↑ resolution
Dark-ground (dark-field)	**Unstained thin organisms – Treponema pallidum (syphilis), Leptospira**; motility
Phase-contrast	Living unstained cells, internal structures
Fluorescence	**Fluorescent dyes – auramine-rhodamine (TB – LED microscopy)**, immunofluorescence (rabies – corneal impression)
Electron microscope (TEM/ SEM)	**Viruses**; **resolution ~0.1–1 nm**; magnification up to ~1,000,000×; **Ernst Ruska & Knoll (1931)**

- **Units** – **1 mm = 1000 µm (micrometre/ micron)**; **1 µm = 1000 nm**; bacteria measured in **µm**, viruses in **nm**.
- **Parts** – eyepiece (ocular), objectives (10×, 40×, **100× oil**), **condenser** (focuses light), iris diaphragm, coarse & fine adjustment, stage; **total magnification = objective × eyepiece**.

1.5 Normal Flora (Microbiota)

Site	Common normal flora
Skin	**Staphylococcus epidermidis (most common)**, S. aureus (nose – 30 % carriers), diphtheroids, Propionibacterium (acne)
Nose & throat	**S. aureus (anterior nares)**, viridans streptococci, Neisseria, H. influenzae
Mouth	**Viridans streptococci (S. mutans – dental caries)**, anaerobes
Stomach	Almost sterile (acid) – H. pylori in some
Large intestine	**Most populated (~10^{11}/g) – Bacteroides (most common, anaerobes), E. coli, Enterococcus, Clostridium, Bifidobacteria, Lactobacilli**
Vagina	**Lactobacilli (Döderlein's bacilli – acidic pH)**, Candida (small numbers)
Urethra	Distal part – S. epidermidis, diphtheroids
Sterile sites	**Blood, CSF, bladder urine, lower respiratory tract, internal organs, peritoneal & pleural cavities**

- **Benefits** – **colonisation resistance (prevent pathogens)**, **vitamin K & B synthesis (gut)**, immune stimulation; **disturbance by antibiotics → superinfection – C. difficile (pseudomembranous colitis), oral/ vaginal candidiasis**.
- **Newborn** – sterile at birth (largely), colonised during delivery & feeding; **breast-fed baby – Lactobacillus bifidus** predominate.

1.6 Importance of Microbiology in Nursing

- **Prevention & control of hospital-acquired infection (HAI)**; asepsis, sterilisation, disinfection.
- **Correct specimen collection, labelling & transport**; interpretation of reports.
- **Safe handling of biomedical waste**; standard precautions; **needle-stick injury management**.
- Immunisation, health education (hand washing, safe water, food hygiene).
- Antibiotic stewardship & AMR awareness; patient isolation.

Yaad karne ki trick

- **"LP-KJ-F"** – **L**eeuwenhoek (saw microbes), **P**asteur (germ theory), **K**och (TB, postulates), **J**enner (vaccine), **F**leming (penicillin), plus **Lister – antisepsis (carbolic)**.
- Koch discovered – **"A-T-C"** – **A**nthrax, **T**B, **C**holera.
- **Pro = no nucleus** ("**pro**blem – no nucleus").

PYQ style

- Father of microbiology – **Antonie van Leeuwenhoek**.
- Father of modern microbiology – **Louis Pasteur**.
- Father of bacteriology – **Robert Koch**.
- Father of antiseptic surgery – **Joseph Lister (carbolic acid)**.
- Father of immunology/ vaccination – **Edward Jenner**.
- Father of chemotherapy – **Paul Ehrlich**.
- Germ theory & pasteurisation – **Pasteur**; rabies vaccine – **Pasteur**.
- TB bacillus discovered – **Robert Koch, 1882**.
- Penicillin – **Alexander Fleming, 1928**.
- Hand washing & puerperal sepsis – **Semmelweis**.
- Gram stain – **Hans Christian Gram**.
- Virus discovered – **Ivanovsky (TMV)**.
- Phagocytosis – **Metchnikoff**.
- Malaria–mosquito link – **Ronald Ross**.
- Prokaryotic ribosome – **70S**; eukaryotic – **80S**.
- Smallest free-living bacteria, no cell wall – **Mycoplasma**.
- Viruses contain – **DNA or RNA (never both)**.
- Prions – **protein only; most resistant**.
- Viruses seen by – **electron microscope**; T. pallidum – **dark-ground microscope**.
- Oil immersion objective – **100×**.
- Commonest skin flora – **Staphylococcus epidermidis**.
- Commonest gut flora – **Bacteroides**.
- Vaginal flora – **Lactobacilli (Döderlein's bacilli)**.

2 Bacterial Morphology, Physiology & Laboratory Methods

2.1 Shape & Arrangement

Bacteria – Shapes, Arrangement & Gram Stain

COCCI (round)

Diplococci

Staphylococci

Streptococci (chain)

BACILLI (rods)

E. coli, Salmonella, TB

Streptobacilli

Club-shaped – Diphtheria

SPIRAL & COMMA

Spirochaetes – Treponema, Leptospira (hooked ends)

Vibrio (comma) – cholera

Size & units

Bacteria ~0.5–5 µm

Virus ~20–300 nm

(smallest – parvo/ polio)

1 µm = 1000 nm

Seen by light microscope

Virus – electron microscope

Gram staining (Hans Christian Gram, 1884) – 4 steps

1. Crystal violet	2. Gram's iodine	3. Acetone/ alcohol	4. Safranin
primary stain – 1 min	mordant – 1 min	decolouriser – KEY step	counterstain

Gram-positive = PURPLE/ violet
thick peptidoglycan wall – retains crystal violet (Staph, Strep, Clostridium, Bacillus, Corynebacterium)

Gram-negative = PINK/ red
thin wall + outer membrane (endotoxin/ LPS) – loses violet (E. coli, Salmonella, Vibrio, Neisseria)

Acid-fast (Ziehl-Neelsen) – TB & leprosy bacilli – RED on blue background

Bacterial shapes, arrangements & Gram staining

Shape	Arrangement	Example
Cocci (spherical)	**Diplococci (pairs)**	**Pneumococcus (lanceolate), Neisseria (kidney/ bean-shaped – gonococcus, meningococcus)**
	Chains	**Streptococci**
	Clusters ("bunch of grapes")	**Staphylococci**
	Tetrads / cubical packets (8)	Micrococcus / **Sarcina**
Bacilli (rods)	Single, chains (streptobacilli)	**E. coli, Salmonella, Bacillus anthracis ("bamboo stick/ box-car")**
	Chinese-letter/ cuneiform, club-shaped	**Corynebacterium diphtheriae**
	Coccobacilli	**Haemophilus, Brucella, Bordetella**
Comma-shaped	–	**Vibrio cholerae** ("fish in stream" – darting motility)
Spiral	Rigid (spirilla) / **flexible (spirochaetes)**	**Treponema (syphilis), Leptospira (hooked ends), Borrelia**

Shape	Arrangement	Example
Pleomorphic	No fixed shape	**Mycoplasma**
Branching filaments	–	**Actinomyces, Nocardia**

2.2 Bacterial Cell Structure

Structure	Features / importance
Cell wall	**Peptidoglycan (murein)** – rigidity & shape; **Gram-positive – thick peptidoglycan (teichoic acid)**; **Gram-negative – thin peptidoglycan + outer membrane with lipopolysaccharide (LPS = endotoxin – lipid A)**; **target of penicillins, cephalosporins, vancomycin**; **lysozyme** (tears, saliva) breaks it; **L-forms/ protoplasts – wall-deficient**; **Mycoplasma – no wall**
Cell (cytoplasmic) membrane	Selective permeability, respiration (no mitochondria), **mesosomes**
Capsule	Polysaccharide (polypeptide in B. anthracis) outer layer – **virulence factor – prevents phagocytosis**; **India ink/ negative staining**; **Quellung reaction (capsule swelling – pneumococcus)**; capsulated – **pneumococcus, Klebsiella, H. influenzae type b, meningococcus, B. anthracis, Cryptococcus (fungus)**
Slime layer/ biofilm	**Adherence to catheters, prosthetic valves, IV lines** (S. epidermidis, Pseudomonas) – antibiotic-resistant
Flagella	**Motility** (protein – flagellin, **H antigen**); types – **monotrichous (one – Vibrio, Pseudomonas)**, **lophotrichous (tuft at one end)**, **amphitrichous (both ends)**, **peritrichous (all over – E. coli, Salmonella, Proteus – swarming)**; **atrichous – non-motile (Shigella, Klebsiella, staph, strep)**; detected by **hanging drop method**
Fimbriae/ pili	**Adhesion (common pili – E. coli in UTI, gonococcus)**; **sex pili – conjugation (transfer of DNA/ plasmids)**
Ribosomes	**70S** – protein synthesis (aminoglycosides, tetracyclines, macrolides target)
Nucleoid	Single circular dsDNA, no nuclear membrane
Plasmids	Extrachromosomal circular DNA – **R plasmids (drug resistance)**, toxin genes; transferred by conjugation
Inclusion granules	**Volutin/ metachromatic granules (Babes-Ernst) – C. diphtheriae (Albert's stain)**; glycogen, lipid
Spores (endospores)	**Highly resistant dormant form** (heat, drying, disinfectants) – **dipicolinic acid + calcium**; **only Gram-positive bacilli – Bacillus (aerobic) & Clostridium (anaerobic)**; **survival form, not reproduction (1 cell → 1 spore)**; position – **central (B. anthracis), subterminal, terminal "drumstick" (C. tetani), oval terminal "tennis racket" (C. tertium)**; **killed by autoclave** – used as **biological indicators** of sterilisation

2.3 Staining Methods

Stain	Type / use
Simple stain	Methylene blue – shape & arrangement

Stain	Type / use
Gram stain	**Differential** – crystal violet (primary) → **Gram's iodine (mordant)** → **acetone/ alcohol (decolouriser – critical step)** → **safranin (counterstain)**; **Gram +ve – violet/ purple; Gram −ve – pink/ red**
Ziehl-Neelsen (acid-fast)	**Carbol fuchsin (heat) → 20 % H_2SO_4 (decolouriser; 5 % for M. leprae) → methylene blue**; **acid-fast bacilli – red/ pink on blue background**; **mycolic acid** in wall; **TB, leprosy, Nocardia (weakly – 1 %), oocysts of Cryptosporidium/ Isospora**
Albert's stain	**Metachromatic granules – C. diphtheriae** (green bacilli, bluish-black granules)
Negative stain (India ink, nigrosin)	**Capsule – Cryptococcus neoformans (CSF)**
Giemsa / Leishman	**Blood smears – malaria parasites**, Leishmania (LD bodies), Chlamydia inclusions
Spore stain	Malachite green (Schaeffer-Fulton)
Fontana (silver impregnation)	Spirochaetes
KOH mount (10–20 %)	**Fungi – skin, nail, hair** (dissolves keratin)
Lactophenol cotton blue	**Fungal culture morphology**
Auramine-rhodamine	Fluorescent – TB

- **RNTCP/ NTEP sputum grading (ZN)** – scanty (1–9 AFB/100 fields), **1+ (10–99/100 fields), 2+ (1–10/field), 3+ (> 10/field)**.

2.4 Bacterial Growth & Physiology

- **Reproduction – binary fission**; **generation (doubling) time** – **E. coli ~20 min**; **M. tuberculosis ~14–18 hrs** (slow – culture takes 4–8 weeks); **M. leprae ~12–14 days** (**can't be cultured in artificial media – armadillo, mouse foot pad**).

Bacterial growth curve (in batch culture)

Phase	Features
1. Lag phase	**Adaptation, no increase in number**, ↑ size & metabolic activity
2. Log (exponential) phase	**Rapid multiplication**; **most susceptible to antibiotics**; ideal for staining/ biochemical tests
3. Stationary phase	Growth = death; nutrient depletion; **spores & exotoxins formed**
4. Decline (death) phase	Death > growth; involution forms

Growth requirements

- **Oxygen** – **obligate aerobes (M. tuberculosis, Pseudomonas, B. subtilis)**; **obligate anaerobes (Clostridium, Bacteroides – killed by O_2; lack catalase/ SOD)**; **facultative anaerobes (most – E. coli, staph)**; **microaerophilic (Campylobacter, H. pylori)**; capnophilic (↑ CO_2 – Brucella, Neisseria).

- **Temperature** – **mesophiles (25–40 °C; most pathogens – optimum 37 °C)**; psychrophiles (cold – Listeria grows at 4 °C – refrigerated food); thermophiles (> 55 °C – Geobacillus).
- **pH** – most 7.2–7.6; **Vibrio cholerae – alkaline (pH 8.2–8.6)**; lactobacilli – acidic.
- **Moisture, nutrients, salt** (S. aureus tolerates 7–10 % NaCl – **halotolerant**; Vibrio parahaemolyticus – halophilic).
- **Anaerobic culture** – **McIntosh & Fildes' jar**, **GasPak**, **Robertson's cooked meat (RCM) medium**, thioglycollate broth.

2.5 Culture Media

Type	Examples
Simple/ basal	**Nutrient broth, nutrient agar, peptone water**
Enriched (+ blood, serum, egg)	**Blood agar (5–10 % sheep blood – haemolysis)**, **chocolate agar (heated blood – Haemophilus, Neisseria)**, Loeffler's serum slope (diphtheria – rapid growth 6–8 hrs)
Enrichment (liquid – favour pathogen, inhibit flora)	**Selenite F broth / tetrathionate (Salmonella from stool)**, **alkaline peptone water (Vibrio)**
Selective (solid)	**Lowenstein-Jensen (LJ – TB)**, **TCBS (Vibrio cholerae – yellow colonies)**, **DCA, XLD, SS agar (Salmonella, Shigella)**, **Thayer-Martin (gonococcus)**, **potassium tellurite (diphtheria – black colonies)**, Wilson-Blair (Salmonella typhi – black metallic sheen), Sabouraud's dextrose agar (**fungi**)
Differential	**MacConkey agar – lactose fermenters (pink – E. coli, Klebsiella) vs non-fermenters (colourless/ pale – Salmonella, Shigella, Proteus)**; blood agar (haemolysis)
Transport media	**Stuart's / Amies (swabs – gonococcus)**, **Cary-Blair, VR medium (Vibrio)**, buffered glycerol saline (stool – Salmonella/ Shigella)
Anaerobic	**Robertson's cooked meat**, thioglycollate
Blood culture	**Brain-heart infusion (BHI) broth, BACTEC/ BacT/ALERT (automated)**; ratio blood : broth **1 : 5 to 1 : 10**
Special	**Agar – solidifying agent (from seaweed; melts at ~90–95 °C, sets at ~40 °C; 1.5–2 %)**; **LJ – egg-based, inspissated (85 °C), malachite green**

- **Haemolysis on blood agar** – **α (partial, greenish – S. pneumoniae, viridans strep)**, **β (complete, clear zone – S. pyogenes, S. aureus)**, **γ (none – Enterococcus)**.
- **Culture methods** – streak plate (isolation of pure colonies), pour plate (colony count), lawn culture (**antibiotic sensitivity – Kirby-Bauer disc diffusion on Mueller-Hinton agar**), stab culture; **MIC** – minimum inhibitory concentration.

2.6 Bacterial Toxins

Feature	Exotoxin	Endotoxin
Source	Secreted by living bacteria – **mostly Gram-positive** (also some Gram −ve – cholera, E. coli LT)	**Part of cell wall (LPS) of Gram-negative** – released on death/ lysis
Chemistry	**Protein**	**Lipopolysaccharide (lipid A)**

Feature	Exotoxin	Endotoxin
Heat	**Heat-labile** (mostly)	**Heat-stable**
Toxicity	**Highly potent**, specific action	Less potent; **fever, shock, DIC** (septic shock)
Toxoid	**Can be converted (formalin) → toxoid – vaccines (tetanus, diphtheria)**	No toxoid
Examples	**Tetanus (tetanospasmin – blocks inhibitory glycine/ GABA release → spasms)**, **botulinum (most potent toxin – blocks ACh release → flaccid paralysis)**, **diphtheria (inhibits protein synthesis – EF-2)**, **cholera (↑ cAMP → rice-water diarrhoea)**, staph enterotoxin, TSST-1, **S. pyogenes (erythrogenic – scarlet fever)**	E. coli, Salmonella, Meningococcus (Waterhouse-Friderichsen)

- **Pyrogen-free (endotoxin-free)** IV fluids; **LAL (Limulus amoebocyte lysate) test** detects endotoxin.

2.7 Bacterial Genetics & Drug Resistance

- **Gene transfer** – **transformation** (uptake of naked DNA – Griffith's pneumococcus experiment), **transduction** (via **bacteriophage**), **conjugation** (**sex pilus – R plasmids, most common mode of spread of multidrug resistance**), transposons ("jumping genes").
- **Mutation** – spontaneous (TB drug resistance → multidrug therapy).
- **Resistance mechanisms** – **enzymatic inactivation (β-lactamase/ ESBL, carbapenemase – NDM-1 "New Delhi metallo-β-lactamase")**, altered target (**MRSA – PBP2a, mecA gene**), ↓ permeability, **efflux pumps**, bypass pathways.
- **Important resistant organisms** – **MRSA, VRE (vancomycin-resistant Enterococcus), ESBL-producing E. coli/ Klebsiella, CRE (carbapenem-resistant Enterobacterales), MDR/ XDR-TB, Acinetobacter baumannii, Pseudomonas**; **WHO priority pathogens list**.
- **Prevention of AMR** – rational antibiotic use (**antimicrobial stewardship**), complete course, **culture & sensitivity before antibiotics**, infection control, hand hygiene, no OTC antibiotics (**Schedule H1, Red Line campaign**), **NAP-AMR (India, 2017)**; **World AMR Awareness Week – 18–24 November**.

Yaad karne ki trick

- Gram stain order – **"Come In And Stain"** – **C**rystal violet, **I**odine, **A**lcohol/ acetone, **S**afranin.
- Spore formers – **"Only B & C make spores"** – Bacillus & Clostridium.
- Capsulated bacteria – **"Some Killers Have Pretty Nice Big Capsules"** – **S**. pneumoniae, **K**lebsiella, **H**. influenzae, **P**seudomonas, **N**. meningitidis, **B**. anthracis, **C**ryptococcus.
- Growth curve – **"LLSD"** – Lag, Log, Stationary, Decline.
- Exotoxin = **Ex**cellent toxoid (vaccine); Endotoxin = **En**d of Gram-negative wall.

PYQ style

- Grape-like clusters – **Staphylococci**; chains – **Streptococci**.
- Kidney-shaped diplococci – **Neisseria**; lanceolate diplococci – **Pneumococcus**.
- Comma-shaped – **Vibrio cholerae**; Chinese-letter – **C. diphtheriae**.
- Bacterial cell wall – **peptidoglycan**; endotoxin – **LPS (Gram-negative)**.
- Capsule function – **prevents phagocytosis**; capsule stain – **India ink**.
- Organ of motility – **flagella**; adhesion – **pili/ fimbriae**; conjugation – **sex pilus**.
- Drug resistance genes carried on – **plasmids (R factor)**.
- Spore-forming bacteria – **Bacillus & Clostridium**; drumstick spore – **C. tetani**.
- Gram mordant – **iodine**; decolouriser – **acetone/ alcohol**; Gram +ve colour – **violet**.
- AFB stain – **Ziehl-Neelsen**; decolouriser – **20 % sulphuric acid**.
- Metachromatic granules – **C. diphtheriae (Albert's stain)**.
- Fungal skin scraping – **10–20 % KOH**.
- Most antibiotic-sensitive phase – **log phase**.
- E. coli generation time – **20 min**; M. tuberculosis – **14–18 hrs**.
- Can't be cultured – **M. leprae, T. pallidum**.
- TB culture medium – **Lowenstein-Jensen**; cholera – **TCBS**; fungi – **Sabouraud's**.
- MacConkey lactose fermenter – **pink colonies (E. coli)**.
- Enriched medium for Haemophilus – **chocolate agar**.
- Anaerobic medium – **Robertson's cooked meat**.
- Antibiotic sensitivity – **Kirby-Bauer on Mueller-Hinton agar**.
- Most potent toxin – **botulinum**; toxoid vaccines – **tetanus, diphtheria**.
- Exotoxin – **protein, heat-labile, toxoid possible**.
- MRSA gene – **mecA**; NDM-1 – **carbapenemase**.

3 Sterilisation & Disinfection

3.1 Definitions

D Definitions

- **Sterilisation** – process that **destroys/ removes ALL microorganisms including bacterial spores**.
- **Disinfection** – destruction of **pathogenic** microbes on **inanimate objects** (not necessarily spores).
- **Antisepsis** – disinfection of **living tissue/ skin** (antiseptic – povidone-iodine, chlorhexidine, alcohol).
- **Asepsis** – absence of pathogenic organisms; **medical asepsis (clean technique)** – reduce number & spread; **surgical asepsis (sterile technique)** – eliminate all microbes.
- **Decontamination** – making items safe to handle (removing/ inactivating pathogens) – **first step**; **cleaning** – physical removal of dirt & organic matter (**must precede disinfection/ sterilisation**).
- **Bactericidal** (kills) vs **bacteriostatic** (inhibits growth); **sporicidal, virucidal, fungicidal**.
- **Sanitisation** – reducing microbes to safe public-health levels (utensils).
- **Fumigation** – disinfection of rooms/ OT with gas/ vapour.

- **Order of resistance (most → least) – prions > bacterial spores > mycobacteria (TB) > non-enveloped (small) viruses (polio, hepatitis A) > fungi > vegetative bacteria > enveloped (lipid) viruses (HIV, HBV, influenza)**.

3.2 Physical Methods

Sterilisation & Disinfection – Methods at a Glance



Linen, gowns, instruments, culture media

Physical methods

- **Hot air oven** (dry heat) – 160 °C × 2 hrs / 180 °C × 30 min – glassware, powders, oils, sharp instruments (Clostridium tetani spores)
- **Incineration** – anatomical waste · **Flaming / red heat** – loops
- **Boiling** 100 °C × 10–30 min – disinfection only (spores survive)
- **Pasteurisation** – milk: 63 °C × 30 min / 72 °C × 15–20 s
- **Tyndallisation** – 100 °C × 20 min, 3 successive days
- **Radiation** – gamma (syringes, sutures – cold sterilisation); UV – surfaces/ air
- **Filtration** – heat-labile fluids (sera, antibiotics); HEPA – air

Chemical methods

- **Ethylene oxide (ETO)** – heat-sensitive (plastics, catheters, heart-lung)
- **2 % glutaraldehyde (Cidex)** – endoscopes: 20 min HLD, 10 hrs sterile

Spaulding classification & common disinfectants

Critical (enter sterile tissue – surgical instruments, needles) → **STERILISATION**
Semi-critical (mucous membrane – endoscopes, laryngoscope) → **HIGH-level disinfection**
Non-critical (intact skin – BP cuff, stethoscope, bedpan) → **LOW-level disinfection**

Hypochlorite 1 % (spills) · 0.5 % (surfaces) · 70 % alcohol · Povidone-iodine · Chlorhexidine (hand/ skin) · Phenol (Lister)

Methods of sterilisation & disinfection with Spaulding classification

Heat (most reliable)

- **Moist heat** – kills by **coagulation & denaturation of proteins**; **more efficient than dry heat** (penetrates better, lower temp & time).
- **Dry heat** – kills by **oxidation**, protein denaturation.

Method	Details	Uses
Autoclave (steam under pressure)	**121 °C, 15 lb/in² (psi), 15–20 min** (most common); **126 °C – 20 psi – 10 min**; **134 °C, 30 psi, 3–5 min (flash/ prevacuum)**; principle – **water boils at higher temp under pressure**; air must be removed (air pockets prevent sterilisation); **most effective method for heat-resistant items**	**Surgical instruments, linen, gowns, dressings, gloves, culture media, lab waste, rubber**
Hot air oven (dry heat)	**160 °C × 2 hrs** (holding time 1 hr in some texts) / **170 °C × 1 hr / 180 °C × 30 min**; fan for circulation; don't overload; **cool before opening (glass breaks)**	**Glassware (syringes, Petri dishes, pipettes), powders (talc), oils, greases, paraffin, sharp metal instruments (no blunting/ rust)**
Incineration	Burning to ash (≥ 800–1050 °C)	**Anatomical waste, soiled dressings, BMW (yellow)**
Red heat/ flaming	Holding in flame till red hot	**Inoculating loops, needles, forceps tips**

Method	Details	Uses
Boiling	**100 °C for 10–30 min – disinfection, NOT sterilisation (spores & some viruses survive)**; add 2 % sodium bicarbonate ↑ efficacy	Feeding bottles, utensils
Pasteurisation	**Holder method – 63 °C × 30 min**; **flash (HTST) – 72 °C × 15–20 s**; (UHT – 135–140 °C × 1–2 s)	**Milk** (kills Brucella, M. bovis, Coxiella, Salmonella)
Tyndallisation (intermittent sterilisation)	**100 °C (steam) × 20 min on 3 successive days** (spores germinate between)	Sugar/ egg/ serum media
Inspissation	**80–85 °C × 30 min on 3 days**	**LJ medium, Loeffler's serum slope**
Vaccine bath	60 °C × 1 hr	Vaccines (killed)

Radiation

- **Ionising (gamma rays – Cobalt-60)** – **"cold sterilisation"** – **disposable syringes, catheters, sutures, gloves, IV sets, bone grafts** (industrial).
- **Non-ionising (UV, 260 nm)** – **poor penetration – surface & air disinfection** (OT, labs, biosafety cabinets); damages eyes & skin.

Filtration

- **Heat-labile fluids** – **sera, antibiotics, vaccines, IV solutions, enzymes**; **membrane filters (0.22 µm)**, Seitz (asbestos), sintered glass, candle filters; **HEPA filters (air – OT, laminar air flow, ICU; remove particles ≥ 0.3 µm with 99.97 % efficiency)**; viruses & mycoplasma may pass.

Others

- **Drying, freeze-drying (lyophilisation – preservation of cultures, vaccines)**, ultrasonic cleaning (instruments), plasma sterilisation (**hydrogen peroxide gas plasma – heat-sensitive items, low temp**).

3.3 Chemical Methods

Agent	Concentration / features	Uses
Alcohols (ethyl, isopropyl)	**60–90 % (70 % best – water needed for protein denaturation)**; **not sporicidal**; quick evaporation	**Skin antisepsis, hand rub, thermometers, surfaces, injection site**
Aldehydes – glutaraldehyde 2 % (Cidex)	**High-level disinfectant (20 min) & sterilant (~10 hrs)**; **sporicidal**; non-corrosive; activated (alkaline) – use within **14 days**; toxic fumes – rinse with sterile water	**Endoscopes, cystoscopes, bronchoscopes, anaesthetic equipment, laparoscopes**
Formaldehyde (formalin – 40 % gas in water)	**10 % formalin – tissue preservation (histopathology)**; gas – **fumigation of OT/ rooms** (with $KMnO_4$ – 500 ml formalin + 250 g $KMnO_4$ per 1000 ft^3; ~6 hrs/ overnight; neutralise with ammonia); carcinogenic, irritant	Fumigation, specimen preservation, toxoid preparation

Agent	Concentration / features	Uses
Ethylene oxide (ETO)	Gas; **alkylating agent – sterilant at low temp (~50–60 °C)**; **toxic, flammable, carcinogenic – needs aeration (12–24 hrs)**	**Heat-sensitive – plastics, catheters, heart-lung machine, cardiac pacemakers, sutures, rubber, optical instruments**
Halogens – chlorine (sodium hypochlorite/ bleach)	**Hypochlorite – 1 % (10,000 ppm available chlorine) for blood/ body-fluid spills & BMW disinfection**; **0.5 % (5000 ppm) – surfaces, linen soaking**; 0.1 % – general; corrosive to metals, inactivated by organic matter; **freshly prepare daily**; **bleaching powder (33 % chlorine)**	Spills, floors, water chlorination (residual 0.5 mg/L), linen
Iodine – povidone-iodine (Betadine 5–10 %), tincture iodine	Iodophor – slow release; **sporicidal (prolonged)**, broad; **allow to dry (2 min)**	**Pre-operative skin prep, wounds, surgical scrub (7.5 %)**
Chlorhexidine	**0.5 % in 70 % alcohol (skin prep – preferred for CVC insertion)**, **2–4 % (hand scrub)**, **0.12–0.2 % mouthwash (VAP prevention)**, **4 % umbilical cord care gel (7.1 % chlorhexidine digluconate)**; **residual action**; not sporicidal	Hand hygiene, skin prep, oral care, catheter care
Phenol & derivatives	**Phenol (carbolic acid) – Lister**; **phenol coefficient (Rideal-Walker, Chick-Martin tests – efficiency vs phenol)**; **cresol (Lysol), chloroxylenol (Dettol), hexachlorophene**	Floors, surfaces, faeces; Dettol – skin
Hydrogen peroxide	**3 % – wound cleaning (effervescence)**; 6–7.5 % – HLD; gas plasma	Wounds, contact lens
Quaternary ammonium compounds (QACs)	**Cetrimide, benzalkonium chloride** – detergent; low-level; inactivated by soap; Savlon = **cetrimide + chlorhexidine**	Surfaces, skin cleaning
Dyes	**Gentian violet (crystal violet)**, acriflavine, brilliant green	Skin, wounds, media (selective)
Heavy metals	**Silver nitrate (1 % – Credé's method for ophthalmia neonatorum – now tetracycline/ erythromycin ointment)**, silver sulfadiazine (burns), mercurials (thiomersal – vaccine preservative)	–
Peracetic acid	Sporicidal, low temp	Endoscopes, dialysers
Potassium permanganate ($KMnO_4$ – 1 : 10,000)	Oxidising; **sitz bath, wound wash, gastric lavage (some poisonings)**, water disinfection of wells (old)	–

- **Factors affecting disinfectant action** – **concentration, contact time, temperature, pH, presence of organic matter (blood, pus ↓ efficacy – clean first!)**, type & number of organisms, nature of surface (biofilm).
- **Ideal disinfectant** – broad-spectrum, fast, not inactivated by organic matter, non-toxic, non-corrosive, stable, cheap, odourless, residual effect.

3.4 Spaulding Classification

Category	Contact	Examples	Process required
Critical	Enter **sterile tissue/ vascular system**	**Surgical instruments, needles, urinary catheters, implants, IV cannula, biopsy forceps**	**Sterilisation**
Semi-critical	**Mucous membranes/ non-intact skin**	**Endoscopes (GI), laryngoscope blades, ET tubes, anaesthesia circuits, vaginal speculum, thermometers (oral/ rectal)**	**High-level disinfection (HLD)** – glutaraldehyde, OPA, peracetic acid, hydrogen peroxide; (boiling/ pasteurisation)
Non-critical	**Intact skin**	**BP cuff, stethoscope, bedpans, crutches, bed rails, linen, furniture**	**Low/ intermediate-level disinfection** – 70 % alcohol, hypochlorite, QACs, phenolics; detergent & water

3.5 Monitoring Sterilisation (Indicators)

Type	Example
Mechanical/ physical	Pressure gauge, thermometer, timer, printouts
Chemical indicators	**Autoclave tape (diagonal stripes turn black)**, **Bowie-Dick test (air removal in pre-vacuum autoclave – daily first run)**, Browne's tubes (red → green), internal indicator strips (class 4–6)
Biological indicators (most reliable – gold standard)	**Autoclave – Geobacillus stearothermophilus spores**; **hot air oven – Bacillus atrophaeus (formerly B. subtilis var. niger) / Clostridium tetani (non-toxigenic) spores**; **ETO – Bacillus atrophaeus**; **ionising radiation – Bacillus pumilus**; filtration – Brevundimonas diminuta / Serratia marcescens

3.6 CSSD & Care of Equipment

- **CSSD (Central Sterile Supply Department)** – zones: **dirty (receiving & decontamination) → clean (inspection, packing) → sterile (storage & issue)** – **unidirectional flow**.
- **Steps for reprocessing** – **decontamination (soaking in 0.5 % hypochlorite × 10 min) → cleaning (detergent, brushing, ultrasonic; disassemble) → drying → inspection → packing (double-wrap, indicator tape, label with date) → sterilisation → storage → issue**.
- **Sterile packs** – shelf life **event-related** (intact, dry, unopened – generally re-sterilise after ~7–30 days per policy in cloth wraps); **wet/ torn/ dropped pack = unsterile**.
- **Sterile field rules** – only sterile touches sterile; **edges (1 inch border) of sterile field are unsterile**; below waist/ table level unsterile; **never turn back on sterile field**; **if in doubt, consider it contaminated**; pour fluids from height without splashing.
- **Rubber goods** – autoclave at lower temp/ time; **gloves – powder before autoclaving**; **catheters – disposable (gamma/ ETO)**.
- **Thermometers** – wash with soap & cold water, wipe with 70 % alcohol **from stem (cleaner) towards bulb (dirtier)**, store dry.
- **Linen** – soiled/ infected – **red bag/ bin**, soak in 0.5–1 % hypochlorite, wash hot (≥ 71 °C × 25 min).
- **Blood/ body fluid spill management** – **wear gloves (PPE) → cover spill with absorbent paper → pour 1 % hypochlorite (10,000 ppm) over it → leave 10–20 min → wipe (outside → inside) →**

discard in yellow bag → clean with detergent; spill kit.

♫ Yaad karne ki trick

- Autoclave – **"121-15-15"** – 121 °C, 15 psi, 15 min.
- Hot air oven – **"160 for 2 hours – glass, powder, oil (dry things)"**.
- Spaulding – **"Critical = Cut (sterile tissue) → Sterilise; Semi = Slimy (mucosa) → HLD; Non = Normal skin → LLD"**.
- Resistance order – **"Prions Spoil My Viral Food Very Easily"** – Prions > Spores > Mycobacteria > non-enveloped Viruses > Fungi > Vegetative bacteria > Enveloped viruses.

✎ PYQ style

- Sterilisation kills – **all microbes including spores**; disinfection – **not necessarily spores**.
- Antiseptic used on – **living tissue**.
- Most resistant infectious agent – **prions**; most resistant bacterial form – **spores**.
- Most reliable sterilisation – **moist heat (autoclave)**.
- Autoclave – **121 °C, 15 psi, 15–20 min**.
- Hot air oven – **160 °C × 2 hrs**; used for – **glassware, powders, oils**.
- Biological indicator for autoclave – **Geobacillus stearothermophilus**.
- Autoclave tape – **chemical indicator**; air removal test – **Bowie-Dick**.
- Pasteurisation (holder) – **63 °C × 30 min**; flash – **72 °C × 15–20 s**.
- Boiling is – **disinfection, not sterilisation**.
- Tyndallisation – **100 °C × 20 min, 3 days**.
- Cold sterilisation – **gamma radiation** (disposable syringes).
- UV rays – **surface & air**; poor penetration.
- Heat-labile fluids – **membrane filtration (0.22 µm)**.
- Endoscope disinfection – **2 % glutaraldehyde**.
- Heat-sensitive plastics, heart-lung machine – **ethylene oxide**.
- OT fumigation – **formalin + $KMnO_4$**.
- Blood spill – **1 % sodium hypochlorite**.
- Best alcohol concentration – **70 %**.
- Phenol coefficient test – **Rideal-Walker**.
- Savlon – **cetrimide + chlorhexidine**.
- Laryngoscope blade – **semi-critical → HLD**.
- Surgical instruments – **critical → sterilisation**.
- BP cuff, stethoscope – **non-critical**.
- First step in reprocessing – **decontamination & cleaning**.

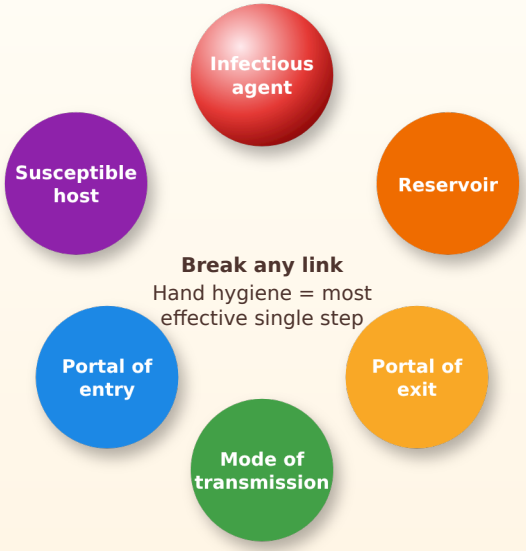
4 Infection, Hospital-Acquired Infections & Infection Control

4.1 Infection – Basic Terms

- **Infection** – entry & multiplication of microorganisms in host tissues; **colonisation** – presence & multiplication **without** tissue invasion/ illness; **contamination** – presence on surface/ object.
- **Types** – **primary, secondary** (on top of primary), **cross-infection** (from another patient/ source), **nosocomial (hospital-acquired)**, **iatrogenic** (due to medical intervention), **subclinical (inapparent)**, **latent** (dormant – HSV, TB), **opportunistic**, **endogenous** (own flora) vs **exogenous**, **zoonosis** (animals → humans – rabies, plague, leptospirosis, brucellosis, anthrax), **focal**, **local/ systemic**.
- **Carrier** – harbours pathogen without disease & can spread it (**typhoid Mary**); **incubatory, convalescent, healthy, chronic (> 1 yr – typhoid in gall bladder)**.
- **Stages of infectious disease** – **incubation period** (entry → first symptoms) → **prodromal** (vague, non-specific) → **illness (acute/ fulminant)** → **decline** → **convalescence**.
- **Epidemic** – sudden ↑ cases above expected in community; **endemic** – constantly present; **pandemic** – worldwide (COVID-19, 1918 flu); **sporadic** – scattered cases; **outbreak** – localised epidemic.

4.2 Chain of Infection & Transmission

Chain of Infection & BMW Colour Coding (Rules 2016)



- Contact (MRSA) · Droplet (> 5 µm – influenza)
- Airborne (< 5 µm – TB, measles – N95, AIIR)
- Vehicle (food, water, blood) · Vector (mosquito, flies)
- Standard precautions for ALL patients



Yellow
Anatomical/ human tissue, placenta, soiled dressings & cotton, expired medicines, cytotoxic, chemical, lab cultures → incineration

Red
Contaminated recyclable plastic – tubing, catheters, IV sets, urine bags, syringes without needles, gloves → autoclave + shred

White (puncture-proof)
Sharps – needles, blades, scalpels

Blue (cardboard box)
Glassware – broken/ discarded glass, vials, ampoules; metallic implants

Untreated waste storage ≤ 48 hrs
Biohazard symbol on bags · Black – general (non-infectious, municipal) waste

Chain of infection & BMW colour coding (BMW Management Rules 2016)

Link	Example	Break by
Infectious agent	Bacteria, virus, fungi, parasites	Early diagnosis, treatment, disinfection, sterilisation
Reservoir	Humans, animals, soil, water, equipment	Environmental cleaning, sanitation, pest control, treat carriers
Portal of exit	Respiratory, GI, blood, skin, urine	Cover cough, dressing wounds, safe disposal of excreta
Mode of transmission	Contact, droplet, airborne, vehicle, vector	**Hand hygiene (most important)**, PPE, isolation, sterile technique
Portal of entry	Broken skin, mucosa, catheters	Asepsis, wound care, catheter care
Susceptible host	Elderly, neonates, immunocompromised, diabetics, malnourished	Immunisation, nutrition, treat underlying disease

Mode	Details	Examples
Contact – direct	Person-to-person (touch, sexual, kissing)	**MRSA, scabies, STIs, impetigo**
Contact – indirect	Via **fomites** (inanimate objects – bed rails, stethoscopes, towels)	**MRSA, C. difficile, rotavirus**
Droplet	**Large particles (> 5 µm), travel ≤ 1–2 m (~3 ft)**, don't remain suspended	**Influenza, diphtheria, pertussis, mumps, rubella, meningococcus, COVID-19 (mainly)**
Airborne	**Droplet nuclei (< 5 µm) – remain suspended, travel long distances**	**TB, measles, varicella (chickenpox), disseminated zoster** ("**My Chicken Has TB**")
Vehicle	Common vehicle – **food, water, blood, IV fluids, drugs**	**Cholera, typhoid, hepatitis A & E, HBV/ HIV (blood)**
Vector	**Mechanical** (flies – typhoid) / **biological** (mosquito – malaria, dengue)	Malaria, dengue, plague, kala-azar
Vertical	Mother → baby (transplacental, during birth, breastfeeding)	**HIV, HBV, syphilis, rubella, toxoplasma, CMV (TORCH)**

4.3 Hospital-Acquired Infections (HAI/ HCAI/ Nosocomial)

- **Definition** – infection **not present or incubating at admission**, appearing **≥ 48 hrs after admission** (or within 3 days of discharge/ **30 days of surgery**, **90 days if implant**).
- **Prevalence** – ~5–10 % of hospitalised patients (higher in ICU).
- **Common organisms** – **Gram-negative bacilli (E. coli, Klebsiella, Pseudomonas, Acinetobacter)**, **S. aureus (MRSA)**, Enterococcus (VRE), CoNS, **Candida**, C. difficile.

HAI	Rank / key prevention ("bundle")
CAUTI (catheter-associated UTI)	**Commonest HAI (~30–40 %)**; **E. coli** commonest; prevention – **avoid unnecessary catheterisation, aseptic insertion, closed drainage system, bag below bladder level (off floor), maintain unobstructed flow, daily review & early removal, perineal hygiene**; don't routinely change catheter/ irrigate

HAI	Rank / key prevention ("bundle")
SSI (surgical site infection)	**S. aureus** commonest; **pre-op bath, clip (don't shave) hair just before surgery, antibiotic prophylaxis within 60 min before incision, glycaemic control, normothermia, OT discipline, aseptic dressing**
VAP (ventilator-associated pneumonia)	**Head end elevation 30–45°**, **daily sedation vacation & readiness-to-wean assessment**, **oral care with chlorhexidine**, subglottic suction, **cuff pressure 20–30 cmH_2O**, DVT & peptic ulcer prophylaxis, hand hygiene
CLABSI (central line-associated BSI)	**Hand hygiene, maximal sterile barrier precautions, chlorhexidine-alcohol skin prep, avoid femoral site (prefer subclavian), daily review & prompt removal**, scrub the hub, transparent dressing changed 7 days (gauze – 2 days)
C. difficile infection	Contact precautions, **soap & water hand wash (alcohol doesn't kill spores)**, hypochlorite cleaning, antibiotic stewardship

4.4 Standard & Transmission-Based Precautions

Standard precautions (CDC 1996) – for ALL patients, all the time

- **Treat all blood, body fluids, secretions, excretions (except sweat), non-intact skin & mucous membranes as potentially infectious**.
- Components – **hand hygiene**, **PPE** (gloves, gown, mask, eye protection as per risk), **respiratory hygiene/ cough etiquette**, **safe injection practices (one needle, one syringe, one time; no recapping)**, **sharps safety**, patient placement, **environmental cleaning**, care of equipment & linen, **waste management**, safe lumbar puncture (mask).
- **Universal precautions (1987 – blood-borne)** were expanded into standard precautions.

Transmission-based	Precautions	Diseases
Contact precautions	**Single room/ cohort**, **gloves & gown on entry**, dedicated equipment (stethoscope, BP cuff)	**MRSA, VRE, C. difficile, scabies, rotavirus, herpes zoster (localised), wound infections, impetigo, RSV**
Droplet precautions	**Single room (or ≥ 1 m/ 3 ft spacing)**, **surgical mask within 1–2 m**, patient wears mask during transport	**Influenza, pertussis, diphtheria (pharyngeal), mumps, rubella, meningococcal meningitis (first 24 hrs of therapy), plague (pneumonic), adenovirus**
Airborne precautions	**AIIR (airborne infection isolation room) – negative pressure, ≥ 6–12 air changes/ hr, door closed**, **N95 respirator (fit-tested)** for staff, **surgical mask for patient** during transport; non-immune staff avoid (measles/ varicella)	**Pulmonary/ laryngeal TB, measles, varicella, disseminated zoster**; (COVID-19 – aerosol-generating procedures)
Protective (reverse) isolation	**Positive-pressure room**, HEPA; protects **immunocompromised** patient (neutropenia, transplant, burns)	–

4.5 Hand Hygiene

- **Single most important & effective measure to prevent HAI**; **"Clean care is safer care" (WHO 2005)**; **World Hand Hygiene Day – 5 May**; **Global Handwashing Day – 15 October**.
- **WHO "My 5 Moments for Hand Hygiene"** – **1. Before touching a patient**; **2. Before clean/ aseptic procedure**; **3. After body fluid exposure risk**; **4. After touching a patient**; **5. After touching patient surroundings**.
- **Alcohol-based hand rub (ABHR)** – **preferred method when hands not visibly soiled**; **20–30 seconds**; ~3 ml.
- **Soap & water hand wash** – **40–60 seconds**; when **hands visibly soiled**, after **toilet**, **C. difficile/ spore-forming organisms, norovirus**, before eating.
- **WHO steps (6–7)** – 1. palm to palm; 2. right palm over left dorsum with interlaced fingers & vice versa; 3. palm to palm fingers interlaced; 4. backs of fingers to opposing palms (fingers interlocked); 5. rotational rubbing of thumbs; 6. rotational rubbing of fingertips in palm; (7. wrists) – then rinse, dry with single-use towel, **turn off tap with towel**.
- **Surgical hand scrub** – **2–5 min (traditionally 5 min first scrub)** with antiseptic (chlorhexidine 4 %/ povidone-iodine 7.5 %) or alcohol-based surgical rub; **hands held above elbows**; fingertips → elbows; **nails short, no rings/ watches/ artificial nails**.
- **Gloves don't replace hand hygiene** – perform before donning & after removing.
- **Most commonly missed areas** – **thumbs, fingertips, between fingers**.

4.6 Personal Protective Equipment (PPE)

- **Donning sequence (CDC)** – **hand hygiene → gown → mask/ respirator → goggles/ face shield → gloves** (gloves over gown cuffs).
- **Doffing sequence** – **gloves (most contaminated – first) → goggles/ face shield → gown → mask/ respirator (last, outside room for N95) → hand hygiene** (alternatively gown + gloves together); **hand hygiene between steps** if contaminated.
- **N95** – filters **≥ 95 % of particles ≥ 0.3 µm**; **fit test & seal check** each use; **surgical mask** – protects from droplets & protects patient from wearer.
- **Gloves** – sterile (surgical procedures), clean/ examination (contact with body fluids); **double gloving** in high-risk surgery; latex allergy – nitrile.

4.7 Needle-Stick Injury (NSI) & Post-Exposure Prophylaxis

- **Risk of transmission after percutaneous exposure** – **HBV ~6–30 % ("rule of 3s": HBV 30 %, HCV 3 %, HIV 0.3 %)**; **HIV via mucous membrane ~0.09 %**.
- **Immediate steps** – **wash wound with soap & running water** (don't scrub, **don't squeeze/ suck**, don't use bleach/ strong antiseptics); **mucosa – flush with water; eye – irrigate with clean water/ saline**; **report immediately to supervisor/ HICC**; document; assess source patient (HIV, HBsAg, HCV status).
- **HIV PEP** – **start as early as possible – ideally within 2 hrs, not later than 72 hrs**; **28 days**; regimen – **TLD (tenofovir + lamivudine + dolutegravir)** (NACO); baseline & follow-up HIV tests (**6 weeks, 3 months, 6 months**).
- **HBV PEP** – if unvaccinated/ non-responder: **HBIG (0.06 ml/kg IM, within 24 hrs, ≤ 7 days) + HBV vaccine series**; vaccinated responder (anti-HBs ≥ 10 mIU/ml) – no treatment.

- **HCV** – no PEP; follow-up testing.
- **Prevention** – **never recap needles (if must – one-handed scoop technique)**, **dispose immediately in puncture-proof sharps container (fill only ¾)**, needle cutters/ destroyers, safety-engineered devices, **HBV vaccination of all HCWs (3 doses – 0, 1, 6 months)**.

4.8 Biomedical Waste Management Rules 2016 (amended 2018, 2019)

Category (bag/ container)	Waste	Treatment/ disposal
Yellow (non-chlorinated plastic bags)	**Human anatomical waste (tissues, organs, body parts, placenta), animal waste, soiled waste (dressings, cotton, plaster casts, bags with blood/ body fluids), expired/ discarded medicines, cytotoxic drugs, chemical waste, discarded linen/ mattresses with body fluids, microbiology lab waste (pre-treat – autoclave on-site), chemical liquid waste**	**Incineration / plasma pyrolysis / deep burial** (rural, where allowed)
Red (non-chlorinated plastic bags/ containers)	**Contaminated recyclable waste – tubing, bottles, IV sets, catheters, urine bags, syringes (without needles & fixed needle syringes with cut needles), vacutainers, gloves**	**Autoclave/ microwave/ hydroclave → shredding → recycling** (not incinerated/ landfilled)
White (translucent) – **puncture-proof, leak-proof, tamper-proof container**	**Sharps – needles, syringes with fixed needles, needles from needle tip cutter/ burner, scalpels, blades**, other contaminated sharp metal	**Autoclave/ dry heat sterilisation → shredding/ encapsulation → iron foundry/ sanitary landfill**
Blue (cardboard boxes with blue marking/ puncture-proof containers)	**Glassware – broken or discarded & contaminated glass, medicine vials & ampoules (except cytotoxic), metallic body implants**	**Disinfection (soaking in hypochlorite)/ autoclave → recycling**

- **Segregation at source (point of generation)** – most important step; **untreated BMW not stored beyond 48 hrs**; **barcoding of bags**; **biohazard & cytotoxic symbols**; **liquid waste – pre-treated (1 % hypochlorite) before drain**; **general (non-infectious) waste – black/ green bags (municipal)**; HCF must have **phase-out of chlorinated plastic bags**; records & annual report to **State Pollution Control Board**.
- **Pre-treatment of lab waste & blood bags – on-site autoclave/ disinfection (as per WHO/ NACO guidelines)** before sending.
- **Mercury spills** – collect with syringe/ cards into airtight container with water (no vacuum cleaner, no broom).

4.9 Infection Control Programme

- **Hospital Infection Control Committee (HICC)** – chairperson (medical superintendent), **infection control officer (microbiologist)**, **infection control nurse (ICN – key person; ~1 per 100–250 beds)**, members from medicine, surgery, nursing, CSSD, housekeeping, pharmacy; meets regularly; policies, surveillance, outbreak investigation, training, antibiotic policy.

- **Surveillance** – HAI rates (per 1000 device-days), **hand hygiene audits**, environmental sampling (OT – air settle plates as indicated), outbreak investigation.
- **Kayakalp** (2015) – award scheme for cleanliness & infection control in public health facilities; **Swachh Bharat**; **NQAS**.
- **Health care worker immunisation** – **hepatitis B (all)**, influenza (annual), varicella/ MMR (non-immune), tetanus, COVID-19.

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- Airborne diseases – **"My Chicken Has TB"** – **M**easles, **Ch**ickenpox (varicella), **H**erpes zoster disseminated, **TB**.
- Droplet – **"PIDMRM"** – **P**ertussis, **I**nfluenza, **D**iphtheria, **M**umps, **R**ubella, **M**eningococcus.
- PPE donning – **"Gown → Mask → Goggles → Gloves" (G-M-G-G)**; doffing – **"Gloves → Goggles → Gown → Mask"** (dirtiest first, mask last).
- NSI risk – **"Rule of 3s" – HBV 30 %, HCV 3 %, HIV 0.3 %**.
- BMW – **"Yellow = burn, Red = recycle plastic, White = sharps, Blue = glass"**.

PYQ style

- Most effective way to prevent HAI – **hand hygiene**.
- WHO hand hygiene moments – **5**; World Hand Hygiene Day – **5 May**.
- Alcohol hand rub duration – **20–30 s**; soap & water – **40–60 s**.
- Soap & water mandatory in – **visibly soiled hands, C. difficile**.
- HAI appears after – **48 hrs of admission**.
- Commonest HAI – **CAUTI (UTI)**; commonest organism – **E. coli**.
- VAP prevention – **head elevation 30–45°, oral chlorhexidine**.
- TB, measles, varicella – **airborne precautions (N95, negative pressure)**.
- Influenza, diphtheria, pertussis – **droplet precautions (surgical mask, 1 m)**.
- MRSA, C. difficile, scabies – **contact precautions**.
- Neutropenic patient – **protective (reverse) isolation – positive pressure**.
- First PPE removed – **gloves**; last – **mask/ respirator**.
- N95 filters – **95 % particles ≥ 0.3 µm**.
- After needle-stick – **wash with soap & water, don't squeeze**.
- HIV PEP – **within 72 hrs (ideally 2 hrs), 28 days**.
- Highest NSI transmission risk – **hepatitis B**.
- Needles – **never recap**; sharps container filled – **¾**.
- Sharps – **white translucent puncture-proof container**.
- Anatomical waste, placenta, soiled dressings – **yellow bag (incineration)**.
- IV tubing, catheters, gloves – **red bag**.
- Glass vials, ampoules – **blue box**.
- Untreated BMW storage – **≤ 48 hrs**.
- BMW rules current – **2016 (amended 2018)**.
- Key person in infection control – **infection control nurse**.

5 Systematic Bacteriology – Important Pathogenic Bacteria

5.1 Gram-Positive Cocci

Organism	Key features	Diseases	Lab/ notes
Staphylococcus aureus	Grape-like clusters; **coagulase +ve**, catalase +ve; **golden-yellow colonies**; β-haemolysis; carried in **anterior nares**	**Boils, furuncles, carbuncles, abscess (pus – commonest pyogenic), impetigo, wound/ SSI**, osteomyelitis (commonest cause), septic arthritis, **food poisoning (preformed enterotoxin – 1–6 hrs, vomiting; milk products)**, **toxic shock syndrome (TSST-1 – tampons)**, **scalded skin syndrome (infants)**, pneumonia (post-influenza), endocarditis (IV drug users – tricuspid), **neonatal sepsis, breast abscess**	**MRSA – vancomycin/ linezolid**; **mupirocin nasal** for carriers
S. epidermidis (CoNS)	Coagulase −ve; **skin flora**; **biofilm**	**Prosthetic devices, IV catheter, CSF shunt infections**; commonest blood-culture contaminant	–
S. saprophyticus	Coagulase −ve, novobiocin-resistant	**UTI in young sexually active women (2nd after E. coli)**	–
Streptococcus pyogenes (Group A β-haemolytic, GAS)	Chains; **bacitracin-sensitive**; M protein	**Pharyngitis/ tonsillitis**, **scarlet fever (erythrogenic toxin – strawberry tongue)**, impetigo, **erysipelas, cellulitis**, necrotising fasciitis, puerperal sepsis; **non-suppurative sequelae – rheumatic fever (after throat infection – 2–3 wks) & acute glomerulonephritis (after throat or skin)**	**ASO titre ↑**; **penicillin DOC**; RF prophylaxis – benzathine penicillin
S. agalactiae (Group B Strep)	Bile-tolerant; vaginal flora	**Neonatal sepsis & meningitis (early onset – commonest in West)**	Antenatal screening 35–37 wks
Viridans streptococci (S. mutans, S. sanguis)	**α-haemolytic**, optochin-resistant, mouth flora	**Dental caries (S. mutans)**, **subacute bacterial endocarditis (damaged valves after dental extraction)**	Antibiotic prophylaxis
Streptococcus pneumoniae (Pneumococcus)	**Lanceolate (flame-shaped) Gram +ve diplococci**, **capsulated**, α-haemolytic, **optochin-sensitive, bile-soluble**; **Quellung reaction**	**Lobar pneumonia (commonest cause of community-acquired pneumonia – rusty sputum)**, **otitis media**, sinusitis, **meningitis (adults – commonest)**, **post-splenectomy sepsis**	**PCV vaccine (UIP)**, PPSV23
Enterococcus faecalis	Group D, gut flora; grows in **6.5 % NaCl, bile esculin**	**UTI, endocarditis, HAI**; **VRE**	Intrinsic resistance

5.2 Gram-Negative Cocci – Neisseria

Organism	Key features	Diseases
N. meningitidis (Meningococcus)	**Kidney/ bean-shaped diplococci (intracellular in neutrophils)**, capsulated; nasopharyngeal carriers; **droplet**	**Meningococcal meningitis (epidemic – cerebrospinal fever; children/ young adults, hostels, Hajj)**, **meningococcaemia – petechial/ purpuric rash**, **Waterhouse-Friderichsen syndrome (adrenal haemorrhage)**; **Rx – penicillin/ ceftriaxone**; **contacts – rifampicin/ ciprofloxacin prophylaxis**; vaccine (A, C, Y, W-135)
N. gonorrhoeae (Gonococcus)	Kidney-shaped diplococci (intracellular); non-capsulated; **Thayer-Martin medium**; **sexual transmission**	**Gonorrhoea – urethritis (purulent discharge – "clap"), cervicitis, PID (infertility, ectopic), ophthalmia neonatorum (onset 2–5 days)**, septic arthritis; **Rx – ceftriaxone (+ azithromycin)**; **no vaccine, no lasting immunity**

5.3 Gram-Positive Bacilli

Organism	Key features	Disease & points
Corynebacterium diphtheriae	**Club-shaped, Chinese-letter arrangement**, **metachromatic (volutin) granules – Albert's stain**; **Loeffler's serum slope** (rapid), **potassium tellurite (black colonies)**; **Elek's gel precipitation test (toxigenicity)**; **Schick test (immunity – historical)**	**Diphtheria – grey-white pseudomembrane (bleeds on removal) on tonsils/ pharynx**, **bull neck**, airway obstruction; **exotoxin → myocarditis (commonest cause of death), neuritis (palatal palsy)**; **Rx – antidiphtheritic serum (ADS – immediately, test dose) + penicillin/ erythromycin**; **droplet**; vaccine – **DPT/ Pentavalent, Td**
Bacillus anthracis	**Largest pathogenic bacterium**, **"bamboo stick/ box-car" chains, central spore, polypeptide capsule (D-glutamic acid)**, non-motile; **"Medusa head" colonies**	**Anthrax – zoonosis (cattle, sheep, goats – wool/ hide)**: **cutaneous (malignant pustule – painless black eschar, commonest)**, **pulmonary (woolsorter's disease – most fatal)**, intestinal; **bioterrorism agent**; Rx – ciprofloxacin/ penicillin/ doxycycline
Bacillus cereus	Spore-former	**Food poisoning – fried rice (emetic type, 1–6 hrs)**
Clostridium tetani	**Obligate anaerobe**, **"drumstick" terminal round spore**, motile; spores in **soil, dust, animal faeces, rusty nails**	**Tetanus – tetanospasmin blocks release of inhibitory neurotransmitters (glycine, GABA) → trismus/ lockjaw (first sign), risus sardonicus, opisthotonus, painful spasms (triggered by light/ noise), dysphagia**; **neonatal tetanus (unclean cord cutting – "8th day disease")**; Rx – **TIG (human tetanus immunoglobulin 3000–6000 IU)/ ATS, wound debridement, metronidazole/ penicillin, diazepam, dark quiet room, airway**; **prevention – TT/ Td vaccine; Td in pregnancy (2 doses/ booster)**

Organism	Key features	Disease & points
Clostridium perfringens (welchii)	Anaerobe, **non-motile**, capsulated, **double zone haemolysis**, **Nagler's reaction (α-toxin – lecithinase)**; stormy fermentation of milk	**Gas gangrene (myonecrosis – crepitus, foul smell, war wounds, crush injury)** – **Rx – debridement, penicillin, hyperbaric O_2**; **food poisoning (meat, 8–24 hrs)**; necrotising enteritis
Clostridium botulinum	Anaerobe, subterminal spore	**Botulism – canned/ preserved food (home-canned), infant botulism (honey) → descending symmetrical flaccid paralysis**, diplopia, ptosis, dysphagia, **no fever, normal sensorium**; **botulinum toxin – most potent toxin known (blocks ACh release)**; Rx – antitoxin; therapeutic Botox
Clostridioides (Clostridium) difficile	Spore-former	**Antibiotic-associated diarrhoea & pseudomembranous colitis (clindamycin, cephalosporins, fluoroquinolones)**; Rx – **oral vancomycin/ fidaxomicin**, (metronidazole); **soap & water hand wash**
Listeria monocytogenes	**Tumbling motility (at 25 °C)**, grows at **4 °C (refrigerator)**	**Neonatal meningitis, listeriosis in pregnancy (abortion)**, unpasteurised cheese; Rx – ampicillin

5.4 Gram-Negative Bacilli – Enterobacteriaceae & Others

Organism	Key features	Diseases & points
Escherichia coli	**Lactose fermenter (pink on MacConkey)**, motile, **commonest gut aerobe**; **indicator of faecal contamination of water**	**UTI (commonest cause – community & hospital)**, **neonatal meningitis (with GBS)**, **diarrhoea** – **ETEC (traveller's diarrhoea; commonest in children of developing countries), EPEC (infantile), EIEC (dysentery-like), EHEC (O157:H7 – haemolytic uraemic syndrome; undercooked beef), EAEC**; sepsis, SSI, **peritonitis**, cholecystitis
Klebsiella pneumoniae	**Lactose fermenter, mucoid colonies (capsule)**, non-motile	**Pneumonia (alcoholics, diabetics – "red currant jelly" sputum, upper lobe cavities)**, UTI, **neonatal sepsis & HAI (ESBL/ carbapenemase – NDM)**
Proteus	**Swarming growth**, NLF, **urease +ve** (alkaline urine), fishy odour	**UTI – struvite (staghorn) renal stones**; Weil-Felix test antigen

Organism	Key features	Diseases & points
Salmonella Typhi & Paratyphi A, B	**NLF, motile (peritrichous)**, H_2S producer; **Vi (capsular), O (somatic), H (flagellar) antigens**; **faeco-oral (food, water, flies); humans only reservoir**; **carriers – gall bladder**	**Enteric (typhoid) fever** – **step-ladder fever, relative bradycardia, coated tongue, rose spots (abdomen, 2nd week), splenomegaly, "pea-soup" diarrhoea or constipation**; complications – **intestinal perforation & haemorrhage (3rd week – Peyer's patches, terminal ileum)**; **diagnosis – "BASU": Blood culture (1st week – best), Antibodies/ Widal (2nd week – rising titre; TO > 1:160 suggestive), Stool culture (3rd week), Urine culture (4th week)**; **bone marrow culture – most sensitive**; Typhidot; **Rx – ceftriaxone, azithromycin (fluoroquinolone resistance common)**; **vaccines – typhoid conjugate vaccine (TCV, ≥ 6 months), Vi polysaccharide (≥ 2 yrs, every 3 yrs), oral Ty21a (live, ≥ 6 yrs)**
Non-typhoidal Salmonella (S. Typhimurium, S. Enteritidis)	Zoonotic – poultry, eggs	**Gastroenteritis/ food poisoning (8–48 hrs)**, bacteraemia, osteomyelitis in sickle cell
Shigella (S. dysenteriae type 1 – most severe, Shiga toxin; **S. flexneri – commonest in India**; S. sonnei)	**Non-motile, NLF**; **very low infective dose (10–100 bacilli)**; faeco-oral (4 Fs – fingers, flies, food, faeces)	**Bacillary dysentery – frequent small stools with blood & mucus, tenesmus, fever, cramps**; **HUS** (S. dysenteriae 1); Rx – **ciprofloxacin/ azithromycin/ ceftriaxone**, ORS
Vibrio cholerae (O1 – classical & El Tor; O139 Bengal)	**Comma-shaped, highly motile (darting – "shooting star"/ "fish in stream"), monotrichous**; **alkaline media – alkaline peptone water (enrichment), TCBS (yellow colonies)**; VR/ Cary-Blair transport; **string test**	**Cholera – cholera toxin ↑ cAMP → profuse painless "rice-water" stools (fishy odour) + vomiting → severe dehydration, hypovolaemic shock, acidosis, hypokalaemia**; **no fever**; incubation **hours–5 days**; **Rx – rehydration (ORS; IV Ringer lactate – best IV fluid), doxycycline 300 mg single dose (adult)/ azithromycin (children, pregnancy)**; **oral cholera vaccine (Shanchol/ Euvichol – 2 doses)**
V. parahaemolyticus	Halophilic	**Seafood food poisoning**
Pseudomonas aeruginosa	**Non-fermenter, strict aerobe, motile, blue-green pigment (pyocyanin, pyoverdin), fruity/ grape-like odour**; **grows in disinfectants, sinks, moist environments**	**Burn wound infections (commonest – green pus)**, **HAI – VAP, CAUTI, ventilators, nebulisers**, **cystic fibrosis lung infection**, **swimmer's ear (otitis externa)**, malignant otitis externa (diabetics), keratitis (contact lens), **ecthyma gangrenosum**; **intrinsically multi-drug resistant** – Rx – piperacillin-tazobactam, ceftazidime, carbapenems, aminoglycosides, colistin
Acinetobacter baumannii	Coccobacilli, non-fermenter	**ICU HAI (VAP, BSI) – MDR/ carbapenem-resistant**

Organism	Key features	Diseases & points
Haemophilus influenzae	**Coccobacilli, needs X (haemin) & V (NAD) factors – "satellitism" around S. aureus on blood agar**; **chocolate agar**; **type b (Hib) – capsulated (polyribosyl ribitol phosphate)**	**Meningitis (children 6 months–2 yrs – before vaccine), epiglottitis (thumb sign on X-ray), pneumonia**, otitis media, conjunctivitis; **Hib vaccine – Pentavalent (UIP)**; contacts – rifampicin
Haemophilus ducreyi	"School of fish"	**Chancroid – soft, painful genital ulcer (soft sore)** + bubo
Bordetella pertussis	**Coccobacilli**, **Bordet-Gengou medium ("mercury drop/ bisected pearl" colonies)**, **cough plate/ pernasal swab**	**Whooping cough (pertussis/ "100-day cough")** – **catarrhal (most infectious) → paroxysmal (whoop, post-tussive vomiting, apnoea in infants) → convalescent**; **lymphocytosis**; complications – pneumonia (commonest cause of death), seizures, sub-conjunctival haemorrhage, hernia; **Rx – azithromycin/ erythromycin**; **DPT/ Pentavalent vaccine, aP in pregnancy (some countries)**
Helicobacter pylori	**Curved, microaerophilic, urease +ve (strongly)**, flagellated	**Chronic gastritis, peptic (duodenal > gastric) ulcer, gastric carcinoma, MALT lymphoma**; **urea breath test, stool antigen, rapid urease test (biopsy)**; Rx – triple therapy (PPI + clarithromycin + amoxicillin)
Campylobacter jejuni	Curved, "seagull wings", microaerophilic, 42 °C	Diarrhoea (poultry); **Guillain-Barré syndrome** association
Yersinia pestis	**Bipolar ("safety-pin") staining**, rat flea (Xenopsylla cheopis)	**Plague – bubonic (commonest), pneumonic (droplet – most infectious), septicaemic**; Rx – streptomycin/ gentamicin, doxycycline; Surat 1994 outbreak
Brucella	Coccobacilli, capnophilic; zoonosis – **unpasteurised milk**, goats, cattle	**Brucellosis (undulant/ Malta fever)**, sacroiliitis, orchitis; occupational (vets, farmers)
Legionella pneumophila	Water systems, air-conditioners, cooling towers	**Legionnaires' disease (atypical pneumonia), Pontiac fever**; urinary antigen

5.5 Mycobacteria

Mycobacterium tuberculosis

- **Robert Koch (24 March 1882)** – **World TB Day 24 March**; **slender, slightly curved, acid-fast bacilli (AFB – mycolic acid)**, **non-motile, non-sporing, non-capsulated**, **obligate aerobe (upper lobes – high O_2)**; **slow growing (generation time ~14–18 hrs; culture 4–8 weeks on LJ medium – rough, buff, "cauliflower/ breadcrumb" colonies)**; **resistant to drying (survives in dried sputum for weeks), killed by sunlight/ UV, boiling, autoclave**; **5 % phenol** for sputum disinfection (or autoclave/ burn).
- **Transmission – airborne droplet nuclei** (cough, sneeze); **M. bovis – unpasteurised milk** (intestinal TB).

- **Pathology** – **Ghon focus** (primary lesion – lower part of upper lobe/ upper part of lower lobe) + hilar lymph nodes = **primary (Ghon) complex**; **caseous necrosis**, granuloma (**Langhans giant cells**, epithelioid cells); **secondary/ reactivation TB – apical cavitation**; **miliary TB** (haematogenous); **extra-pulmonary – lymph node (commonest – scrofula), pleural, abdominal, spine (Pott's spine – thoracic), meningitis (children), genitourinary, bone**.
- **Diagnosis** – **sputum smear microscopy (ZN/ fluorescent – 2 samples, spot + early morning)**; **CBNAAT/ GeneXpert (MTB + rifampicin resistance in 2 hrs) & TrueNat – now first-line (NTEP)**; **culture (LJ – gold standard; liquid MGIT – faster 10–14 days)**; **LPA (line probe assay – INH & RIF resistance)**; chest X-ray; **Mantoux (TST) – 5 TU PPD, 0.1 ml intradermal, read at 48–72 hrs, ≥ 10 mm induration positive (≥ 5 mm in HIV)** – type IV hypersensitivity, indicates infection not disease; **IGRA (QuantiFERON)**; ADA (pleural/ CSF).
- **BCG vaccine** – live attenuated **M. bovis (Bacille Calmette-Guérin, Danish 1331 strain in India)**; **at birth (up to 1 yr), 0.05 ml (< 1 month)/ 0.1 ml, intradermal, left upper arm (insertion of deltoid)**; reconstituted with **normal saline**, use within **4 hrs**; protects mainly against **severe childhood TB (miliary, meningitis)**; **scar** forms in 6–12 weeks.
- **Treatment (NTEP)** – **2HRZE + 4HRE (daily, FDC by weight)**; **DOTS / 99DOTS**; **MDR-TB** (resistant to H & R) – BPaLM/ shorter oral bedaquiline regimens; **XDR-TB**; **Ni-kshay portal; Ni-kshay Poshan Yojana**; **TB Mukt Bharat target** (End TB by 2025 in India – SDG 2030 globally).

Mycobacterium leprae

- **Gerhard Armauer Hansen (1873)** – **Hansen's disease**; **acid-fast (weakly – 5 % H_2SO_4 decolouriser)**, **"cigar bundles"/ globi** in macrophages; **can't be cultured in artificial media – mouse foot pad, nine-banded armadillo**; **longest generation time (~12–14 days)**; **incubation 3–5 yrs (long)**; **prefers cooler body parts (skin, peripheral nerves, nose, testes, ear lobes)**; transmission – **nasal droplets** (main), prolonged close contact; **man – only reservoir**.
- **Classification (WHO)** – **paucibacillary (PB – 1–5 skin lesions, smear −ve)** & **multibacillary (MB – ≥ 6 lesions, nerve involvement > 1, smear +ve)**; **Ridley-Jopling – TT, BT, BB, BL, LL** (tuberculoid – good CMI; lepromatous – poor CMI, many bacilli, **leonine facies**, madarosis, saddle nose).
- **Cardinal signs** – **hypopigmented/ erythematous anaesthetic skin patch**, **thickened/ tender peripheral nerves (ulnar – commonest, common peroneal, great auricular)**, **slit-skin smear +ve**.
- **Lepromin test** – prognostic (not diagnostic) – +ve in tuberculoid.
- **Treatment – MDT (WHO 1982)**: **rifampicin (monthly supervised) + dapsone (daily) + clofazimine (monthly + daily)** – **PB 6 months, MB 12 months** (uniform 3-drug for both since 2018); **lepra reactions (type 1 – reversal; type 2 – ENL)**; **NLEP – Sparsh Leprosy Awareness Campaign (30 January – Anti-Leprosy Day)**.
- **Atypical mycobacteria (NTM)** – M. avium complex (HIV), M. marinum (fish tank granuloma), M. ulcerans (Buruli ulcer), M. fortuitum (post-surgical, injection abscess).

5.6 Spirochaetes

Organism	Disease & points
Treponema pallidum	**Syphilis** – **sexual, transplacental (congenital – after 16 weeks), blood**; **can't be cultured**; seen by **dark-ground microscopy**; **primary – painless hard chancre (indurated ulcer, ~3 weeks)** → **secondary (rash on palms & soles, condylomata lata, mucous patches, lymphadenopathy – most infectious)** → latent → **tertiary (gumma, cardiovascular – aortitis, neurosyphilis – tabes dorsalis, Argyll Robertson pupil)**; **congenital – Hutchinson's teeth, saddle nose, snuffles, sabre tibia, interstitial keratitis, 8th nerve deafness (Hutchinson's triad)**; **tests – non-treponemal screening VDRL/ RPR (titres – monitor response; false +ve in pregnancy, SLE, leprosy, malaria)**, **treponemal confirmatory – TPHA, FTA-ABS** (remain +ve lifelong); **Rx – benzathine penicillin G 2.4 MU IM** (Jarisch-Herxheimer reaction); **antenatal VDRL screening (eliminate congenital syphilis)**
Leptospira interrogans	**Leptospirosis (Weil's disease)** – **zoonosis – rat urine** contaminated water/ soil (**floods, paddy fields, farmers, sewer workers**); enters via abraded skin/ mucosa; **fever, myalgia (calf tenderness), conjunctival suffusion**, jaundice, renal failure, pulmonary haemorrhage; **hooked ends, dark-ground**; **MAT (microscopic agglutination test – gold standard)**, IgM ELISA; **Rx – doxycycline/ penicillin**; **chemoprophylaxis – doxycycline 200 mg weekly** (high-risk exposure); common in **Gujarat (south – Surat, Valsad, Navsari)**, Kerala, Maharashtra
Borrelia	B. recurrentis – **relapsing fever (louse-borne)**; B. burgdorferi – **Lyme disease (tick – erythema migrans)**

5.7 Rickettsia, Chlamydia & Mycoplasma

Organism	Features & diseases
Rickettsia (obligate intracellular, vector-borne)	**Epidemic typhus (R. prowazekii – louse)**, endemic/ murine typhus (rat flea), **Indian tick typhus (R. conorii)**, **Rocky Mountain spotted fever**; **scrub typhus (Orientia tsutsugamushi – larval trombiculid mite "chigger"; eschar – painless black scab)**; **Q fever (Coxiella burnetii – no vector, inhalation; not rickettsia now)**; **Weil-Felix test (Proteus OX19, OX2, OXK – heterophile)**; **Rx – doxycycline (DOC)**; rash, fever, headache
Chlamydia trachomatis (obligate intracellular – elementary & reticulate bodies; inclusion bodies – Halberstaedter-Prowazek)	**Trachoma (serotypes A–C) – leading infectious cause of blindness** (follicles, Herbert's pits, trichiasis, entropion; **SAFE strategy – Surgery, Antibiotics (azithromycin), Facial cleanliness, Environmental improvement**); **genital (D–K) – commonest bacterial STI, non-gonococcal urethritis, cervicitis, PID, infertility, ectopic; inclusion conjunctivitis & pneumonia in newborn (5–14 days)**; **lymphogranuloma venereum (L1–L3)**; Rx – **azithromycin/ doxycycline**
Chlamydophila pneumoniae / psittaci	Atypical pneumonia; **psittacosis (parrots/ birds)**
Mycoplasma pneumoniae	**Smallest free-living organism, no cell wall (penicillin & cephalosporins ineffective)**, pleomorphic, **"fried egg" colonies**; **primary atypical pneumonia ("walking pneumonia") – young adults**; **cold agglutinins (IgM)**; Rx – **macrolides/ doxycycline**; M. hominis, Ureaplasma – genital

Yaad karne ki trick

- Typhoid diagnosis by week – **"BASU"** – **B**lood (1st), **A**ntibody/ Widal (2nd), **S**tool (3rd), **U**rine (4th).
- Tetanus signs – **"Lockjaw → Smile (risus) → Arch (opisthotonus)"**.
- Syphilis congenital – **Hutchinson's triad – teeth, keratitis, deafness**.
- TB drugs side effects – **"RIPE"** – Rifampicin (orange), INH (neuropathy), Pyrazinamide (gout), Ethambutol (eyes).
- Leprosy nerve – **"Ulnar is Usual"**.

PYQ style

- Commonest cause of pyogenic infections/ boils – **Staphylococcus aureus**.
- Coagulase-positive – **S. aureus**; prosthetic device infections – **S. epidermidis**.
- Staph food poisoning – **preformed toxin, 1–6 hrs**.
- Rheumatic fever follows – **Group A Strep throat infection**; marker – **ASO titre**.
- Dental caries – **Streptococcus mutans**; SABE – **viridans streptococci**.
- Lobar pneumonia commonest – **Streptococcus pneumoniae**.
- Kidney-shaped diplococci – **Neisseria**; meningitis epidemics – **meningococcus**.
- Ophthalmia neonatorum (2–5 days) – **gonococcus**.
- Diphtheria – **pseudomembrane**; commonest cause of death – **myocarditis**.
- Largest pathogenic bacteria – **Bacillus anthracis**; woolsorter's disease – **pulmonary anthrax**.
- Tetanus first sign – **trismus (lockjaw)**; toxin – **tetanospasmin**.
- Gas gangrene – **Clostridium perfringens**.
- Botulism – **descending flaccid paralysis; canned food**.
- Pseudomembranous colitis – **Clostridioides difficile**.
- Commonest cause of UTI – **E. coli**; indicator of faecal pollution – **E. coli**.
- Typhoid – 1st week test – **blood culture**; Widal – **2nd week**; perforation – **3rd week**.
- Carrier site in typhoid – **gall bladder**.
- Rice-water stools – **cholera**; transport medium – **VR/ Cary-Blair**; selective – **TCBS**.
- Best IV fluid in cholera – **Ringer lactate**.
- Bacillary dysentery – **Shigella**.
- Burn wound green pus – **Pseudomonas aeruginosa**.
- X & V factors – **Haemophilus influenzae**.
- Whooping cough – **Bordetella pertussis**; medium – **Bordet-Gengou**.
- Peptic ulcer – **H. pylori (urease)**.
- Plague vector – **rat flea (Xenopsylla)**.
- TB discovered – **Koch 1882**; World TB Day – **24 March**.
- Mantoux positive – **≥ 10 mm induration at 48–72 hrs**.
- BCG – **live attenuated M. bovis, intradermal, left upper arm**.
- Rapid TB + rifampicin resistance – **CBNAAT (GeneXpert)**.
- Leprosy bacillus – **Hansen**; cultured in – **armadillo/ mouse foot pad**.
- Commonest nerve in leprosy – **ulnar**.
- Syphilis – **Treponema pallidum**; screening – **VDRL**; Rx – **benzathine penicillin**.
- Leptospirosis – **rat urine**; sign – **conjunctival suffusion, calf tenderness**.
- Scrub typhus – **mite; eschar; doxycycline**.
- Trachoma – **Chlamydia trachomatis; SAFE strategy**.
- No cell wall – **Mycoplasma**; cold agglutinins – **M. pneumoniae**.

6 Virology

6.1 General Properties of Viruses

- **Smallest infectious agents (20–300 nm)**; **acellular**, **obligate intracellular parasites**; contain **only one type of nucleic acid (DNA or RNA)**; **no ribosomes, no energy system**; **not affected by antibiotics**; pass through bacterial filters; seen by **electron microscope** (**largest – poxvirus ~300 nm, visible by light microscopy – Paschen bodies; smallest – parvovirus ~20 nm, picornaviruses ~25–30 nm**).
- **Structure** – **virion** = **nucleic acid core + capsid (protein coat made of capsomeres)** = **nucleocapsid**; ± **envelope** (lipid, from host membrane – **enveloped viruses sensitive to ether, detergents, alcohol, drying**); **peplomers/ spikes** (glycoproteins – haemagglutinin, neuraminidase).
- **Symmetry** – **icosahedral (cubic – adenovirus, polio, herpes)**, **helical (influenza, measles, rabies – bullet-shaped)**, **complex (poxvirus – brick-shaped, bacteriophage – tadpole)**.
- **Replication steps** – **adsorption (attachment) → penetration → uncoating → biosynthesis (eclipse phase – no infectious virus detectable) → assembly (maturation) → release (lysis or budding)**.
- **Retroviruses (HIV)** – **reverse transcriptase (RNA → DNA)**, integrates into host genome (provirus).
- **Cultivation** – **animal inoculation (suckling mice)**, **embryonated hen's egg** (chorioallantoic membrane – pox "pocks", amniotic & allantoic cavity – influenza, yolk sac – Chlamydia/ Rickettsia), **tissue/ cell culture (most common – cytopathic effect CPE)**.
- **Inclusion bodies** (diagnostic) – **Negri bodies (rabies – hippocampus, Purkinje cells – intracytoplasmic)**, **Guarnieri/ Paschen bodies (vaccinia/ smallpox)**, **owl's eye (CMV – intranuclear)**, **Cowdry type A (herpes – intranuclear)**, **molluscum bodies**, **Warthin-Finkeldey giant cells (measles)**.
- **Interferons** – host proteins (**IFN-α, β, γ**) produced by virus-infected cells – **non-specific antiviral** (block replication in neighbouring cells); used therapeutically (hepatitis B/C – older, MS).
- **Oncogenic viruses** – **HPV (cervical cancer – types 16, 18)**, **HBV & HCV (hepatocellular carcinoma)**, **EBV (Burkitt's lymphoma, nasopharyngeal carcinoma)**, **HHV-8 (Kaposi's sarcoma)**, **HTLV-1 (adult T-cell leukaemia)**.
- **Bacteriophage** – virus infecting bacteria (phage typing, transduction).

6.2 DNA Viruses

Family / virus	Disease & points
Herpesviridae (enveloped, dsDNA, **latency**)	**HSV-1 – oral (cold sores/ herpes labialis), encephalitis (temporal lobe – commonest sporadic viral encephalitis), keratitis**; **HSV-2 – genital herpes (painful vesicles/ ulcers), neonatal herpes (delivery – LSCS if active lesions)**; latency in **sensory ganglia (trigeminal/ sacral)**; **Tzanck smear – multinucleated giant cells**; Rx – **acyclovir**
Varicella-zoster virus (VZV, HHV-3)	**Chickenpox – centripetal rash (trunk), pleomorphic (macule → papule → vesicle "dewdrop on rose petal" → crust – all stages at once), itchy; airborne + contact; infectious 1–2 days before rash till all lesions crusted**; incubation **14–16 days (10–21)**; complications – secondary bacterial infection (commonest), pneumonia (adults), encephalitis, **congenital varicella syndrome**, **Reye's (aspirin)**; **herpes zoster (shingles) – reactivation in dorsal root ganglion → painful dermatomal vesicular rash, unilateral; post-herpetic neuralgia**; vaccine (live); VZIG

Family / virus	Disease & points
Cytomegalovirus (CMV, HHV-5)	**Commonest congenital viral infection – deafness (sensorineural), microcephaly, periventricular calcification, chorioretinitis**; **transplant/ HIV (retinitis)**; owl's eye inclusions; Rx – ganciclovir
Epstein-Barr virus (EBV, HHV-4)	**Infectious mononucleosis ("kissing disease") – fever, pharyngitis, lymphadenopathy, splenomegaly, atypical lymphocytes; Paul-Bunnell/ Monospot (heterophile antibody) test**; **ampicillin rash**; Burkitt's lymphoma, nasopharyngeal carcinoma
HHV-6 / 7	**Roseola infantum (exanthem subitum – high fever 3–5 days, rash appears as fever subsides)**
HHV-8	**Kaposi's sarcoma (AIDS)**
Hepatitis B virus (HBV – Hepadnaviridae; partially dsDNA, reverse transcriptase)	**Dane particle (complete virion, 42 nm)**; transmission – **blood, sexual, vertical (perinatal – highest chronicity ~90 % in newborn)**, needle-stick; **incubation 30–180 days (avg ~75 days, 6 weeks–6 months)**; **markers – HBsAg (first marker; > 6 months = chronic carrier), HBeAg (high infectivity/ active replication), anti-HBc IgM (acute infection – "window period" marker), anti-HBs (immunity – recovery or vaccination; protective ≥ 10 mIU/ml)**, HBV DNA; complications – **chronic hepatitis, cirrhosis, hepatocellular carcinoma**; **vaccine – recombinant HBsAg (yeast) – birth dose within 24 hrs + Pentavalent at 6, 10, 14 wks**; adults – **0, 1, 6 months (deltoid IM; not gluteal)**; **newborn of HBsAg +ve mother – HBIG 0.5 ml + vaccine within 12–24 hrs (different thighs)**; **survives 7 days on surfaces; 100 times more infectious than HIV**; Rx – tenofovir, entecavir
Human papillomavirus (HPV – Papillomaviridae)	**Warts (common, plantar), genital warts – condyloma acuminata (types 6, 11)**, **cervical cancer (types 16, 18 – ~70 %)**, oropharyngeal, anal cancers; **koilocytes**; **HPV vaccine – girls 9–14 yrs (2 doses; single dose now acceptable per WHO); India's indigenous quadrivalent vaccine Cervavac (2022)**; **Pap smear/ VIA screening**
Poxviridae	**Smallpox (variola – eradicated 1977 last natural case – Ali Maow Maalin, Somalia; declared eradicated 8 May 1980; India – last case 1975; vaccine – vaccinia, Jenner)**; **molluscum contagiosum (pearly umbilicated papules)**; **mpox (monkeypox) – PHEIC 2022 & 2024**
Adenoviridae	**Pharyngoconjunctival fever, epidemic keratoconjunctivitis ("pink eye" – swimming pools), acute respiratory infection, gastroenteritis (types 40, 41), haemorrhagic cystitis**
Parvovirus B19 (smallest DNA virus, ssDNA)	**Erythema infectiosum (fifth disease – "slapped cheek" rash)**, **aplastic crisis in sickle cell/ haemolytic anaemia**, **hydrops fetalis**

6.3 RNA Viruses

Virus	Disease & points
Poliovirus (Picornaviridae – Enterovirus; types 1, 2, 3)	**Faeco-oral**; multiplies in gut → viraemia → **anterior horn cells of spinal cord** → **acute flaccid paralysis (asymmetric, lower limbs, no sensory loss)**; **~90–95 % inapparent, < 1 % paralytic**; **provocation polio (IM injections)**; **India – last wild case 13 January 2011 (Howrah, WB); WHO certified polio-free 27 March 2014**; **WPV2 eradicated 2015, WPV3 2019**; **cVDPV** risk; **vaccines – OPV (live, Sabin – bOPV types 1 & 3; at birth, 6, 10, 14 wks, 16–24 months), IPV (killed, Salk – fIPV intradermal 0.1 ml at 6 & 14 wks + 9 months)**; **AFP surveillance – any child < 15 yrs with AFP – 2 stool samples 24 hrs apart within 14 days, sent in reverse cold chain**; **Pulse Polio (1995)**
Coxsackie, echovirus, enterovirus 71	**Hand-foot-mouth disease (Coxsackie A16, EV71)**, **herpangina**, **myocarditis/ pericarditis, Bornholm disease (pleurodynia – Coxsackie B)**, aseptic meningitis
Hepatitis A virus (HAV – Picornavirus)	**Faeco-oral (water, food)**; **incubation 15–45 days (avg ~30)**; children – mild/ anicteric; **no chronicity/ carrier state**; **anti-HAV IgM – diagnosis**; vaccine (killed/ live)
Hepatitis E virus (HEV)	**Faeco-oral (water-borne epidemics – Delhi 1955–56)**; **incubation 2–9 weeks (~40 days)**; **high mortality in pregnancy (~20–25 %, 3rd trimester – fulminant hepatitis)**; no chronicity (except immunocompromised)
Hepatitis C virus (HCV – Flaviviridae)	**Blood-borne (transfusion, IV drug use, dialysis, unsafe injections)**; **most likely to become chronic (~70–85 %)** → cirrhosis, HCC; **no vaccine**; **curable with DAAs (sofosbuvir + velpatasvir/ daclatasvir, 12 weeks)**; **National Viral Hepatitis Control Programme (2018)**
Hepatitis D virus (Delta)	**Defective RNA virus – needs HBsAg (HBV) to replicate**; **co-infection or super-infection (more severe)**; HBV vaccine protects
Rotavirus (Reoviridae – double-stranded RNA, "wheel-like")	**Commonest cause of severe diarrhoea (dehydrating gastroenteritis) in infants & young children (6–24 months)**, winter; **faeco-oral**; **Rotavac (indigenous) vaccine – oral, 6, 10, 14 wks (UIP since 2016)**
Norovirus (Caliciviridae)	**Outbreaks of gastroenteritis (cruise ships, hospitals, "winter vomiting disease")**; hand washing with soap
Measles (rubeola – Paramyxoviridae, Morbillivirus)	**Airborne, highly infectious (R_0 12–18)**; **incubation ~10 days (to fever), 14 days to rash**; **prodrome – 3 Cs: cough, coryza, conjunctivitis + high fever**; **Koplik's spots (buccal mucosa opposite lower molars – pathognomonic, appear 1–2 days before rash)**; **maculopapular rash starts behind ears/ hairline → face → trunk → limbs (day 4), fades with brownish staining & branny desquamation**; infectious **4 days before to 4 days after rash**; complications – **pneumonia (commonest cause of death)**, **diarrhoea**, otitis media, encephalitis, **SSPE (years later)**, **corneal ulceration & blindness (vitamin A deficiency)**, malnutrition; **vitamin A supplementation** (age-based doses × 2 days); **MR vaccine – 9–12 months & 16–24 months (UIP)**; **live attenuated (Edmonston-Zagreb)**; reconstitute with diluent, **use within 4 hrs**, protect from light
Mumps (Paramyxovirus)	**Droplet & saliva**; **incubation ~16–18 days**; **parotitis (bilateral, painful – ear lobe lifted)**; complications – **orchitis (post-pubertal males – sterility rare)**, oophoritis, **aseptic meningitis (commonest extra-salivary)**, pancreatitis, deafness; MMR vaccine

Virus	Disease & points
Rubella (German measles – Togaviridae)	**Mild fever, rash (face → body, 3 days – "3-day measles"), post-auricular & suboccipital lymphadenopathy (characteristic)**; **congenital rubella syndrome (CRS) – 1st trimester infection: classic triad – cataract (eye), PDA/ cardiac defects, sensorineural deafness (most common)** + microcephaly, IUGR, "blueberry muffin" rash; **MR/ MMR vaccine; avoid pregnancy for 1 month after vaccination**
Influenza (Orthomyxoviridae – segmented RNA)	**Haemagglutinin (H – attachment) & neuraminidase (N – release)**; **antigenic drift (minor – seasonal epidemics, A & B)** vs **antigenic shift (major – reassortment, only type A → pandemics: 1918 H1N1 Spanish flu, 2009 H1N1 swine flu)**; **avian flu H5N1**; droplet; **Rx – oseltamivir (within 48 hrs)**; annual vaccine
RSV (respiratory syncytial virus – Paramyxoviridae)	**Commonest cause of bronchiolitis & pneumonia in infants < 1–2 yrs**; syncytia; nirsevimab/ palivizumab (prophylaxis)
Rabies virus (Rhabdoviridae – bullet-shaped, Lyssavirus)	**Zoonosis – dog bite (~96–97 % in India)**, cats, monkeys, mongoose, jackals, bats; **travels along nerves (retrograde axonal) to CNS**; **incubation 1–3 months (variable – shorter in face/ head/ hand bites)**; **hydrophobia (fear of water – spasm of pharyngeal muscles), aerophobia, hypersalivation, agitation → paralysis → death (~100 % fatal once symptomatic)**; **Negri bodies**; **diagnosis – corneal impression smear (IFA), skin biopsy (nape of neck), saliva PCR**; **WHO categories – I (licks on intact skin – no PEP), II (minor scratches, nibbling – vaccine), III (transdermal bites, licks on broken skin/ mucosa, bat exposure – vaccine + RIG)**; **wound wash with soap & running water for 15 min (most important first aid)**, antiseptic, **don't suture immediately**; **ARV – intramuscular Essen (days 0, 3, 7, 14, 28) or Zagreb 2-1-1 (2 doses on day 0, 1 each on days 7 & 21)/ intradermal Updated Thai Red Cross (0.1 ml at 2 sites on days 0, 3, 7, 28)**; **RIG (ERIG 40 IU/kg, HRIG 20 IU/kg) infiltrated into & around wound**; **deltoid/ anterolateral thigh – never gluteal**; **pre-exposure – days 0, 7 (21/28)**; **World Rabies Day – 28 September (Pasteur's death anniversary)**; **NRCP – Zero by 30** (dog-mediated rabies elimination by 2030)
Dengue (Flaviviridae – 4 serotypes DENV 1–4)	**Vector – Aedes aegypti (day-biter, breeds in clean stagnant water – coolers, tyres, pots; "tiger mosquito")**; **incubation 4–7 days**; **dengue fever – "breakbone fever": high fever, retro-orbital pain, myalgia, arthralgia, rash**, **thrombocytopenia**; **warning signs (abdominal pain, persistent vomiting, mucosal bleed, lethargy, liver enlargement, rising haematocrit with falling platelets)**; **severe dengue (DHF/ DSS) – plasma leakage – secondary infection with different serotype (antibody-dependent enhancement)**; **critical phase – defervescence (day 3–7)**; **tourniquet test**; **NS1 antigen (day 1–5), IgM ELISA (after day 5)**; **Rx – paracetamol, fluids; avoid aspirin/ NSAIDs/ steroids**; **National Dengue Day – 16 May**; no vaccine in UIP (Qdenga/ Dengvaxia abroad)
Chikungunya (Togaviridae – Alphavirus)	**Aedes**; **severe, prolonged polyarthralgia ("that which bends up")**, fever, rash; rarely fatal
Japanese encephalitis (Flavivirus)	**Culex (tritaeniorhynchus) – rice fields**; **pigs – amplifier hosts**, ardeid birds – reservoir; **man – dead-end host**; children; high mortality & neurological sequelae; **JE vaccine (live SA 14-14-2) – endemic districts at 9 & 16–24 months**
Zika (Flavivirus)	Aedes; **microcephaly (congenital)**, GBS
Yellow fever (Flavivirus)	Aedes; Africa & S. America; **not in India**; **vaccine (17D) – valid lifelong; certificate for travellers – 10 days after vaccination**; India requires certificate/ quarantine (6 days) for arrivals from endemic areas

Virus	Disease & points
KFD (Kyasanur forest disease – Flavivirus)	**Ticks (Haemaphysalis); monkeys; Karnataka ("monkey fever")**
Nipah virus (Paramyxoviridae, Henipavirus)	**Fruit bats (Pteropus) – reservoir**; pigs; **date palm sap**; **encephalitis & respiratory illness, high fatality (40–75 %)**; outbreaks – **Kerala (2018, 2019, 2021, 2023…), West Bengal (2001, 2007)**; no vaccine
SARS-CoV-2 (Coronaviridae – crown-like spikes)	**COVID-19 – pandemic (declared 11 March 2020; PHEIC ended May 2023)**; **droplet/ aerosol & contact**; **spike protein – ACE2 receptor**; **incubation 2–14 days (~5)**; anosmia, ageusia, fever, cough; **RT-PCR (gold standard), rapid antigen test**; variants – Delta, Omicron; India vaccines – **Covishield (viral vector), Covaxin (inactivated, indigenous), ZyCoV-D (DNA)**; SARS (2002), MERS (camels, 2012)
HIV (Retroviridae – Lentivirus; ssRNA, reverse transcriptase)	See 6.4
Ebola, Marburg (Filoviridae)	**Viral haemorrhagic fever**; contact with body fluids; very high mortality
Hantavirus	Rodents; HFRS, pulmonary syndrome

6.4 HIV/ AIDS

- **HIV-1 (worldwide, more virulent; most common in India) & HIV-2 (West Africa)**; **retrovirus** – 2 copies of ssRNA, **reverse transcriptase, integrase, protease**; envelope glycoproteins **gp120 (binds CD4) & gp41**; core **p24**.
- **Target cells – CD4+ T-helper lymphocytes** (also macrophages, dendritic cells, microglia) via **CD4 + co-receptors CCR5/ CXCR4**.
- **Transmission** – **sexual (commonest – ~85–90 % in India)**, **blood & blood products**, **sharing needles (IDU)**, **mother-to-child (transplacental, intrapartum – maximum, breastfeeding)** – **MTCT risk ~30–45 % without intervention, < 2 % with ART**; **NOT spread by** casual contact, hugging, sharing utensils, mosquitoes, swimming pools.
- **Inactivated by** – heat (56 °C × 30 min), 70 % alcohol, 1 % hypochlorite (enveloped – fragile outside body).
- **Natural history** – **acute HIV (2–4 wks – seroconversion illness, flu-like; high viral load, infectious)** → **window period (infected but antibody −ve – ~3–12 weeks; 4th gen tests reduce it)** → **clinical latency (asymptomatic, 8–10 yrs)** → **symptomatic** → **AIDS (CD4 < 200 cells/µl or AIDS-defining illness)**; **normal CD4 – 500–1500/µl**.
- **Opportunistic infections (OIs)** – **TB (commonest OI & commonest cause of death in India)**, **oral candidiasis (commonest fungal)**, **Pneumocystis jirovecii pneumonia (PCP – CD4 < 200; cotrimoxazole prophylaxis)**, **cryptococcal meningitis (CD4 < 100)**, **toxoplasmosis (brain abscess)**, **CMV retinitis (CD4 < 50)**, MAC, cryptosporidiosis/ isosporiasis (chronic diarrhoea), herpes zoster, **Kaposi's sarcoma**, lymphoma, HIV wasting, PML.
- **Diagnosis (NACO)** – **ELISA/ rapid tests (3 different antigens/ principles – "strategy III" for diagnosis)**; **p24 antigen (early)**; **HIV DNA PCR – infants < 18 months (maternal antibodies interfere – Early Infant Diagnosis at 6 weeks)**; **Western blot** (older confirmatory); **viral load (RNA PCR) – monitoring**; **CD4 count**.

- **ART** – **"Test & Treat" (all HIV+ regardless of CD4, since 2017)**; first-line **TLD (tenofovir + lamivudine + dolutegravir)**; adherence > 95 %; **PPTCT – lifelong ART to pregnant women + infant nevirapine prophylaxis (6–12 weeks)**; **PEP – within 72 hrs, 28 days**.
- **National programmes** – **NACO (1992)**, **NACP-V (2021–26)**; **ICTC** (Integrated Counselling & Testing Centre); **HIV/ AIDS (Prevention & Control) Act 2017**; **World AIDS Day – 1 December**; **red ribbon**; **UNAIDS 95-95-95 targets**.

♫ Yaad karne ki trick

- Measles prodrome – **"3 Cs" – Cough, Coryza, Conjunctivitis** + Koplik's spots.
- CRS triad – **"Cataract, Cardiac (PDA), Cochlear (deafness)"**.
- Hepatitis faeco-oral – **"A & E – vowels – from the bowels"**; blood-borne – **B, C, D**.
- Most chronic hepatitis – **"C = Chronic"**.
- Rabies categories – **"I – Intact skin (nothing), II – vaccine, III – vaccine + RIG"**.
- Herpes latency – **HSV-1 trigeminal, HSV-2 sacral, VZV dorsal root ganglia**.

PYQ style

- Viruses contain – **DNA or RNA**; antibiotics – **ineffective**.
- Largest virus – **poxvirus**; smallest DNA virus – **parvovirus**.
- Negri bodies – **rabies**; owl's eye – **CMV**; Koplik's spots – **measles**.
- Antiviral host protein – **interferon**.
- Cervical cancer – **HPV 16, 18**; Burkitt's lymphoma – **EBV**.
- Chickenpox rash – **centripetal, pleomorphic**; shingles – **VZV reactivation**.
- Commonest congenital viral infection – **CMV**.
- Kissing disease – **EBV (infectious mononucleosis)**.
- Slapped cheek – **parvovirus B19**.
- HBV first marker – **HBsAg**; infectivity – **HBeAg**; window period – **anti-HBc IgM**; immunity – **anti-HBs**.
- HBV vaccine adult schedule – **0, 1, 6 months, deltoid**.
- Newborn of HBsAg+ mother – **HBIG + vaccine within 12–24 hrs**.
- Hepatitis with high mortality in pregnancy – **hepatitis E**.
- Defective virus needing HBV – **hepatitis D**.
- Most chronic – **hepatitis C**.
- Severe infant diarrhoea – **rotavirus**.
- India polio-free certified – **27 March 2014**; last case – **13 January 2011**.
- AFP stool samples – **2 samples, 24 hrs apart, within 14 days**.
- Measles commonest complication/ death – **pneumonia**.
- Mumps commonest complication – **aseptic meningitis**; post-pubertal – **orchitis**.
- CRS commonest defect – **deafness**.
- Pandemic influenza due to – **antigenic shift**.
- Bronchiolitis in infants – **RSV**.
- Rabies first aid – **wash wound with soap & water for 15 min**.
- Category III bite – **vaccine + RIG**.
- Dengue vector – **Aedes aegypti (day biter)**; test day 1–5 – **NS1 antigen**.
- JE vector – **Culex**; amplifier – **pigs**.
- Nipah reservoir – **fruit bats**.
- HIV target – **CD4 T cells**; enzyme – **reverse transcriptase**.
- AIDS defined by CD4 – **< 200/µl**.
- Commonest OI in Indian HIV patients – **TB**.
- HIV diagnosis in infant < 18 months – **DNA PCR**.
- World AIDS Day – **1 December**; World Rabies Day – **28 September**.

7 Mycology & Parasitology

7.1 Fungi – General

- **Eukaryotic**, **cell wall of chitin**, cell membrane contains **ergosterol** (target of **amphotericin B – binds ergosterol; azoles – inhibit its synthesis**); heterotrophic; reproduce by **spores (sexual & asexual)**.
- **Morphological types** – **yeasts (unicellular, budding – Cryptococcus)**, **yeast-like (pseudohyphae – Candida)**, **moulds/ filamentous (hyphae → mycelium – Aspergillus, Mucor, dermatophytes)**, **dimorphic (yeast at 37 °C in tissue, mould at 25 °C – Histoplasma, Blastomyces, Sporothrix, Talaromyces/ Penicillium marneffei)** ("**Mould in the cold, yeast in the heat**").
- **Hyphae** – **septate (Aspergillus – acute-angle/ 45° dichotomous branching)** vs **aseptate/ pauciseptate, broad, ribbon-like, right-angle branching (Mucor/ Rhizopus – mucormycosis)**.
- **Lab diagnosis** – **KOH mount (10–20 % – skin, nail, hair)**, **India ink (Cryptococcus capsule – CSF)**, **Gram stain (Candida – Gram +ve budding yeast)**, **culture – Sabouraud's dextrose agar (SDA, with chloramphenicol/ cycloheximide)**, **lactophenol cotton blue mount**, **germ tube test (Candida albicans – within 2 hrs)**, **Wood's lamp (UV – Microsporum fluoresces green; tinea versicolor yellow-gold)**, serology (cryptococcal antigen, galactomannan – Aspergillus, β-D-glucan).

7.2 Fungal Infections (Mycoses)

Type	Disease / organism	Points
Superficial	**Pityriasis (tinea) versicolor – Malassezia furfur**	**Hypo/ hyperpigmented scaly patches (trunk) – "spaghetti & meatballs" on KOH**; selenium sulphide, azoles
	Tinea nigra, piedra	–
Cutaneous – dermatophytes (Trichophyton, Microsporum, Epidermophyton)	**Tinea (ringworm)** – **capitis (scalp – children; griseofulvin)**, **corporis (body – ring-shaped with central clearing, active edge)**, **cruris (groin – "jock itch")**, **pedis (athlete's foot)**, **unguium (onychomycosis – nails; terbinafine)**, barbae, faciei	Keratin-loving; **KOH mount**; **topical azoles/ terbinafine; oral itraconazole/ terbinafine/ griseofulvin**; keep dry, don't share towels/ combs; **Epidermophyton – skin & nails, not hair**; **Microsporum – skin & hair, not nails**
Mucocutaneous – Candida albicans	**Oral thrush (white curd-like patches that can be scraped – infants, steroids/ inhaler, HIV, diabetics, antibiotics)**, **vulvovaginal candidiasis (curdy white discharge, itching – pregnancy, diabetes, OCP, antibiotics)**, **intertrigo, napkin (diaper) rash (satellite lesions)**, paronychia, oesophagitis (HIV), **candidaemia (IV lines, ICU)**	**Commonest opportunistic fungal infection**; normal flora; **germ tube +ve**; Rx – **nystatin, clotrimazole, fluconazole**; **C. auris – MDR, outbreaks in ICUs**

Type	Disease / organism	Points
Subcutaneous	**Mycetoma (Madura foot – Madurella; actinomycetoma – Nocardia) – swelling, sinuses, grains/ granules**; **sporotrichosis ("rose gardener's disease" – lymphatic spread)**; **chromoblastomycosis**; **rhinosporidiosis (Rhinosporidium seeberi – nasal polyps; ponds; Tamil Nadu)**	Trauma (thorn prick, barefoot); India – common
Systemic (endemic, dimorphic)	**Histoplasmosis (bird/ bat droppings – caves)**, blastomycosis, coccidioidomycosis, paracoccidioidomycosis, **talaromycosis (Manipur – bamboo rats, HIV)**	Inhalation; lungs
Opportunistic	**Cryptococcus neoformans (pigeon droppings) – meningitis in HIV (CD4 < 100) – India ink capsule, CrAg**; **Aspergillus fumigatus – aspergilloma ("fungal ball" in TB cavity), ABPA, invasive aspergillosis (neutropenia)**; **Mucormycosis (Rhizopus, Mucor) – rhino-orbito-cerebral ("black fungus") in uncontrolled diabetes/ DKA, steroids – post-COVID epidemic 2021 India**; **Pneumocystis jirovecii – PCP in HIV (cotrimoxazole)**	Rx – **amphotericin B (liposomal – DOC for mucormycosis & cryptococcal meningitis)**, voriconazole (aspergillosis), surgical debridement

- **Mycotoxins** – **aflatoxin (Aspergillus flavus – groundnuts, maize stored in humid conditions → hepatocellular carcinoma)**; **ergotism (Claviceps purpurea – rye)**; **epidemic dropsy (argemone oil – not fungal)**.

7.3 Parasitology – General

- **Parasite** – lives on/ in host, deriving benefit; **ectoparasite** (lice, mites – scabies) vs **endoparasite**.
- **Definitive host** – harbours **adult/ sexual stage**; **intermediate host** – **larval/ asexual stage**; **reservoir host**; **paratenic**; **man as definitive host** for most helminths; **man as intermediate host – malaria (mosquito is definitive), hydatid disease (dog definitive), Toxoplasma (cat definitive), cysticercosis (Taenia solium larva)**.
- **Protozoa** – unicellular; **trophozoite (active, feeding, motile) & cyst (resistant, infective form – most intestinal protozoa)**; **helminths** – multicellular worms – **nematodes (roundworms), cestodes (tapeworms), trematodes (flukes)**.
- **Lab** – **stool examination – saline mount (trophozoites, eggs) & iodine mount (cysts – nuclei)**; **formol-ether concentration**; **3 consecutive samples**; **stool for trophozoites – warm & fresh, within 30 min**; **peripheral blood smear (thick – detection, thin – species)**, rapid antigen tests.

7.4 Protozoa

Organism	Disease & points
Entamoeba histolytica	**Amoebiasis** – **faeco-oral (cysts – quadrinucleate, infective form)**; **trophozoite with ingested RBCs (diagnostic – pathogenic)**; **amoebic dysentery (stools with blood & mucus, foul smell, less frequent than bacillary; flask-shaped ulcers in caecum/ colon)**; **amoebic liver abscess (commonest extra-intestinal; right lobe; "anchovy sauce" pus, sterile)**; Rx – **metronidazole/ tinidazole + luminal agent (diloxanide furoate/ paromomycin)**
Giardia lamblia (intestinalis)	**First protozoan seen (Leeuwenhoek)**; **pear/ kite-shaped trophozoite, "falling leaf" motility, 2 nuclei – "face/ old man with spectacles"**; **duodenum & jejunum**; **steatorrhoea, malabsorption, foul-smelling bulky greasy stools, flatulence**; children; Rx – **metronidazole/ tinidazole**
Trichomonas vaginalis	**Sexually transmitted**; **no cyst stage (trophozoite infective)**; **frothy greenish-yellow foul-smelling discharge, "strawberry cervix"**, itching; **jerky/ twitching motility (wet mount)**; **treat partner** – metronidazole 2 g single dose/ 500 mg BD × 7 days
Plasmodium	**Malaria** – see 7.5
Leishmania donovani	**Kala-azar (visceral leishmaniasis – "black fever")** – **vector: sandfly (Phlebotomus argentipes)**; **LD (Leishman-Donovan) bodies – amastigotes in macrophages (spleen, bone marrow aspirate)**; **fever, massive splenomegaly, hepatomegaly, pancytopenia, darkening of skin, weight loss**; **rK39 rapid test**; **Bihar, Jharkhand, WB, UP**; **Rx – single-dose liposomal amphotericin B (10 mg/kg)**, miltefosine; **PKDL** (post-kala-azar dermal leishmaniasis); **elimination target – < 1 case/ 10,000 population at block level**
L. tropica	**Cutaneous leishmaniasis (oriental sore)** – Rajasthan
Toxoplasma gondii	**Cat – definitive host** (oocysts in cat faeces); humans – **undercooked meat (tissue cysts), cat litter, transplacental**; **congenital toxoplasmosis – chorioretinitis, hydrocephalus, intracranial calcifications (diffuse), convulsions (triad)**; **brain abscess in HIV (ring-enhancing lesions)**; pregnant women – avoid cat litter, undercooked meat; Rx – pyrimethamine + sulfadiazine; spiramycin (pregnancy)
Cryptosporidium parvum, Cystoisospora (Isospora) belli, Cyclospora	**Chronic watery diarrhoea in HIV/ AIDS**; **modified acid-fast stain (oocysts)**; chlorine-resistant (Cryptosporidium – water outbreaks)
Balantidium coli	**Largest protozoan & only ciliate pathogenic to humans**; pigs; dysentery
Trypanosoma	**T. brucei – African sleeping sickness (tsetse fly)**; **T. cruzi – Chagas disease (reduviid/ kissing bug – S. America)**
Naegleria fowleri	**"Brain-eating amoeba" – primary amoebic meningoencephalitis (swimming in warm fresh water – enters via cribriform plate)** – Kerala cases
Acanthamoeba	Keratitis (contact lens), granulomatous amoebic encephalitis

7.5 Malaria

- **Species** – **P. vivax (commonest in India overall ~ half, relapses)**, **P. falciparum (most dangerous – cerebral malaria, severe/ complicated; ~ half in India, dominant in Odisha, NE, tribal/ forest**

areas), P. malariae (quartan, nephrotic syndrome), P. ovale (relapses; Africa), **P. knowlesi (monkeys, SE Asia)**.

- **Vector – female Anopheles mosquito** (bites at **night/ dusk–dawn**; breeds in **clean water**); **India – An. culicifacies (rural – main vector ~65 %), An. stephensi (urban – overhead tanks, wells), An. fluviatilis (hills/ foothills – streams), An. minimus & An. dirus (NE – forest), An. sundaicus (coastal – brackish water)**.
- **Life cycle** – **mosquito = definitive host (sexual cycle – sporogony)**; **man = intermediate host (asexual cycle – schizogony)**; **infective form to man – sporozoites (mosquito saliva)**; **infective form to mosquito – gametocytes**; **exo-erythrocytic (liver) cycle → merozoites → erythrocytic cycle** (ring forms → trophozoites → schizonts → RBC rupture → fever paroxysm); **hypnozoites (dormant liver forms) – P. vivax & P. ovale → relapses (primaquine kills)**; **extrinsic incubation period in mosquito ~10–14 days** (temperature dependent); **intrinsic incubation (man) – vivax ~14 days, falciparum ~12 days**.
- **Fever periodicity** – **tertian (every 48 hrs) – vivax, ovale ("benign tertian"), falciparum ("malignant tertian" – often irregular)**; **quartan (72 hrs) – malariae**; **quotidian (24 hrs) – knowlesi/ mixed**.
- **Paroxysm stages** – **cold stage (chills, rigors – 15–60 min) → hot stage (high fever 2–6 hrs) → sweating stage (2–4 hrs)**.
- **Falciparum features** – **all ages of RBCs infected (high parasitaemia), multiple ring forms per RBC, accolé/ appliqué forms, banana/ crescent-shaped gametocytes (diagnostic), schizonts not seen in peripheral blood (sequestration in capillaries – cytoadherence)**; **complications – cerebral malaria (coma), severe anaemia, hypoglycaemia, ARDS, AKI (blackwater fever – haemoglobinuria), shock (algid malaria), jaundice, DIC**; **vivax – Schüffner's dots, enlarged RBCs, amoeboid trophozoites**.
- **Diagnosis** – **peripheral blood smear (gold standard) – thick smear (detection, more sensitive), thin smear (species identification)** – **Giemsa/ JSB/ Leishman stain**; **RDT (bivalent – HRP-2 for Pf, pLDH for Pv) – ASHA level, 15–20 min**; QBC; PCR; **slide collection before antimalarials, from finger prick (3rd/ 4th finger) of all fever cases**.
- **Treatment (NCVBDC/ NVBDCP, 2013 & updates)** – **vivax: chloroquine 25 mg/kg over 3 days (10-10-5) + primaquine 0.25 mg/kg × 14 days (radical cure; G6PD; not in pregnancy & infants)**; **falciparum: ACT – artesunate (3 days) + sulfadoxine-pyrimethamine (day 1) [AS + SP] – rest of India; artemether-lumefantrine (AL) – North-East states** + **primaquine 0.75 mg/kg single dose (gametocidal, day 2)**; **severe malaria – injectable artesunate (IV/ IM)**, quinine (alternative); **1st trimester pregnancy Pf – quinine** (ACT now also allowed per WHO).
- **Prevention** – **LLINs (long-lasting insecticidal nets)**, **IRS (indoor residual spraying – DDT/ synthetic pyrethroids/ malathion)**, larvicides (temephos/ Abate), **larvivorous fish (Gambusia, guppy)**, source reduction, repellents; chemoprophylaxis (travellers – doxycycline 100 mg daily/ mefloquine weekly); **RTS,S & R21 vaccines (WHO – Africa)**.
- **National Framework for Malaria Elimination (2016–2030)** – **zero indigenous cases by 2027, elimination by 2030**; **World Malaria Day – 25 April**; **Anti-Malaria Month – June**.
- **Indicators** – **API (annual parasite incidence = confirmed cases/ population × 1000)**, **ABER (annual blood examination rate ≥ 10 %)**, SPR, SFR.

7.6 Helminths

Nematodes (roundworms)

Worm	Points
Ascaris lumbricoides (roundworm)	**Largest intestinal nematode (female 20–35 cm)**; **faeco-oral (embryonated eggs – soil)**; **larval lung migration → Loeffler's syndrome (cough, eosinophilia)**; **intestinal obstruction (worm bolus) in children**, biliary obstruction, malnutrition; **albendazole 400 mg**
Hookworm (Ancylostoma duodenale, Necator americanus)	**Filariform larva penetrates skin (walking barefoot – feet) – "ground itch"**; attach to **jejunum** – suck blood (**~0.2 ml/day per worm – A. duodenale**) → **iron deficiency (microcytic hypochromic) anaemia**, hypoproteinaemia; **eosinophilia**; prevention – **wear footwear, sanitary latrines**; albendazole + iron
Enterobius vermicularis (pinworm/ threadworm)	**Commonest in children**; **perianal itching at night (female lays eggs at anus)**; **autoinfection** (scratching – fingernails); **NIH swab/ cellophane (Scotch) tape test – early morning before bath** (eggs D-shaped/ planoconvex); **treat whole family**; albendazole repeat after 2 weeks; cut nails, wash hands
Trichuris trichiura (whipworm)	**Barrel/ lemon-shaped eggs with bipolar plugs**; caecum; **rectal prolapse**, dysentery, anaemia
Strongyloides stercoralis	**Skin penetration; autoinfection → hyperinfection syndrome (steroids, HTLV-1)**; **larvae (rhabditiform) in stool**; ivermectin
Wuchereria bancrofti, Brugia malayi (filaria)	**Lymphatic filariasis** – **vector: Culex quinquefasciatus (W. bancrofti – urban, dirty water)**, **Mansonia (Brugia – water plants, Kerala)**; **adult worms in lymphatics → lymphoedema, elephantiasis (legs), hydrocele (commonest chronic manifestation in men)**, chyluria, **tropical pulmonary eosinophilia**; **microfilariae – nocturnal periodicity (10 PM–2 AM – night blood smear)**; **ICT card test (antigen)**; **Rx/ MDA – DEC 6 mg/kg + albendazole (DA); IDA (ivermectin + DEC + albendazole – triple drug) in selected districts**; **MMDP (morbidity management) – foot hygiene for lymphoedema**; elimination target **2027**
Dracunculus medinensis (Guinea worm)	**Drinking water from step-wells containing Cyclops (water flea)**; worm emerges from blister on leg; **India certified free in 2000**; prevention – filtering water through cloth
Toxocara	Visceral larva migrans (dog/ cat)
Trichinella spiralis	Undercooked pork – myositis, periorbital oedema, eosinophilia
Ancylostoma braziliense	**Cutaneous larva migrans (creeping eruption)** – dog/ cat hookworm

Cestodes (tapeworms) – flat, segmented, hermaphrodite

Worm	Points
Taenia solium (pork tapeworm)	**Undercooked pork (cysticerci) → intestinal taeniasis**; **ingestion of eggs (faeco-oral/ autoinfection) → cysticercosis – neurocysticercosis (commonest cause of acquired epilepsy in developing countries; "starry sky" calcifications)** – man acts as intermediate host; **pig – intermediate host**; Rx – **albendazole/ praziquantel + steroids**

Worm	Points
Taenia saginata (beef tapeworm)	Undercooked beef; **longest tapeworm (up to 10 m)**; no cysticercosis in man
Echinococcus granulosus (dog tapeworm)	**Hydatid disease** – **dog = definitive host; sheep/ man = intermediate host (accidental)**; **liver (commonest ~60–70 %, right lobe) > lungs**; **hydatid cyst – "hydatid sand", daughter cysts**; **rupture → anaphylaxis**; **Casoni test (old)**, USG/ CT, serology; **Rx – surgery (PAIR) + albendazole**
Hymenolepis nana (dwarf tapeworm)	**Smallest tapeworm; commonest tapeworm in humans (children)**; no intermediate host needed (direct, autoinfection)
Diphyllobothrium latum (fish tapeworm)	**Largest tapeworm** (up to 10–25 m); **vitamin B12 deficiency (megaloblastic anaemia)**; raw fish

Trematodes (flukes) – flat, unsegmented, leaf-shaped; snail intermediate host

- **Schistosoma haematobium** – **urinary schistosomiasis (terminal haematuria, bladder cancer – squamous)**; **S. mansoni, S. japonicum** – intestinal/ hepatic (portal hypertension); **cercariae penetrate skin in fresh water (swimmer's itch)**; praziquantel.
- **Fasciola hepatica (sheep liver fluke)**, **Fasciolopsis buski (largest intestinal fluke – water plants)**, **Clonorchis sinensis (Chinese liver fluke – cholangiocarcinoma, raw fish)**, **Paragonimus westermani (lung fluke – crabs; haemoptysis mimics TB – NE India)**.

7.7 Important Vectors (Medical Entomology)

Vector	Diseases
Anopheles (female)	**Malaria**, (filaria in some regions)
Aedes aegypti/ albopictus	**Dengue, chikungunya, Zika, yellow fever**
Culex	**Filaria (Bancroftian), Japanese encephalitis, West Nile**
Mansonia	**Brugian filaria**
Sandfly (Phlebotomus)	**Kala-azar**, oriental sore, sandfly fever
Housefly (Musca)	**Mechanical – typhoid, cholera, dysentery, diarrhoea, trachoma, polio, amoebiasis**
Rat flea (Xenopsylla cheopis)	**Plague, endemic (murine) typhus**
Body louse (Pediculus)	**Epidemic typhus, relapsing fever, trench fever**
Ticks	**KFD, Indian tick typhus, Lyme disease, CCHF (Crimean-Congo haemorrhagic fever – Gujarat outbreaks)**, relapsing fever
Trombiculid mite (chigger)	**Scrub typhus**
Itch mite (Sarcoptes scabiei)	**Scabies (burrows – finger webs, wrists; nocturnal itch; permethrin 5 % to all family)**
Cyclops	**Guinea worm**, fish tapeworm
Reduviid bug	Chagas disease
Tsetse fly	Sleeping sickness

Vector	Diseases
Snail	Schistosomiasis, flukes

♫ Yaad karne ki trick

- Dimorphic fungi – **"Mould in the Cold, Yeast in the Heat (body)"**.
- Malaria – **"Mosquito = Master (definitive), Man = Middle (intermediate)"**.
- Relapsing malaria – **"V & O – Vivax & Ovale – hypnozoites"**.
- Pinworm test – **"Scotch tape in the morning"**.
- Hydatid – **"Dog Defines, Sheep & man Suffer"**.
- Aedes – **"ADZY"** – Aedes: Dengue, Zika, Yellow fever (+ chikungunya).

✎ PYQ style

- Fungal cell membrane sterol – **ergosterol**; cell wall – **chitin**.
- Fungal culture medium – **Sabouraud's dextrose agar**; skin scraping – **KOH**.
- Cryptococcus capsule – **India ink**; source – **pigeon droppings**.
- Oral thrush – **Candida albicans**; germ tube test – **C. albicans**.
- Ringworm – **dermatophytes**; scalp ringworm – **griseofulvin**.
- Black fungus – **mucormycosis (diabetes, steroids)**; DOC – **liposomal amphotericin B**.
- Aflatoxin – **Aspergillus flavus → liver cancer**.
- Amoebic liver abscess pus – **anchovy sauce**; infective form – **quadrinucleate cyst**.
- Steatorrhoea & "falling leaf" motility – **Giardia**.
- Frothy discharge, strawberry cervix – **Trichomonas vaginalis**.
- Kala-azar vector – **sandfly**; LD bodies – **Leishmania**; Rx – **single-dose L-AmB**.
- Toxoplasma definitive host – **cat**.
- Malaria vector – **female Anopheles**; definitive host – **mosquito**.
- Infective form to man – **sporozoite**; to mosquito – **gametocyte**.
- Relapse – **hypnozoites (vivax, ovale) – primaquine**.
- Most dangerous – **P. falciparum**; banana-shaped gametocytes – **falciparum**.
- Gold standard malaria diagnosis – **peripheral blood smear**.
- Pv treatment – **chloroquine 3 days + primaquine 14 days**.
- Severe malaria – **injectable artesunate**.
- World Malaria Day – **25 April**.
- Largest intestinal nematode – **Ascaris**; Loeffler's syndrome – **Ascaris larvae**.
- Hookworm – **iron deficiency anaemia; enters through skin**.
- Perianal itching in children – **Enterobius (pinworm)**; test – **cellophane tape**.
- Filaria vector – **Culex**; microfilaria periodicity – **nocturnal**; commonest chronic sign in men – **hydrocele**.
- Guinea worm – **Cyclops; India free in 2000**.
- Neurocysticercosis – **Taenia solium eggs**.
- Hydatid cyst commonest site – **liver**; definitive host – **dog**.
- Vitamin B12 deficiency – **Diphyllobothrium latum**.
- Scrub typhus – **mite**; plague – **rat flea**; kala-azar – **sandfly**.

8 Immunity, Serology & Vaccines

8.1 Immunity – Overview

Type	Components	Features
Innate (natural, non-specific)	**Skin & mucosa (1st line), lysozyme (tears, saliva), gastric acid, cilia, normal flora**; **phagocytes (neutrophils, macrophages), NK cells, complement, inflammation, fever, interferons (2nd line)**	Present from birth, rapid, **no memory**
Acquired (adaptive, specific – 3rd line)	**B cells → antibodies (humoral)**; **T cells (cell-mediated)**	**Specificity, memory, diversity**; slower first time (primary response)

Acquired	Active (own immune system; lasting; lag period)	Passive (ready-made antibodies; immediate; short ~weeks–months)
Natural	**Clinical/ subclinical infection**	**Transplacental IgG, breast milk/ colostrum IgA**
Artificial	**Vaccines (live, killed, toxoid, subunit)**	**Immunoglobulins & antisera – ATS/ TIG, ADS, ASV, RIG, HBIG, VZIG, anti-D**

- **Herd immunity** – immunity of a population (enough immune individuals interrupt transmission); threshold depends on R_0 (**measles ~95 %**, polio ~80–85 %).
- **Primary immune response** – after first exposure; lag ~7–10 days; **mainly IgM**, low titre; **secondary (anamnestic/ booster) response** – faster, higher, longer, **mainly IgG** (basis of booster doses).
- **Antigen** – substance inducing specific immune response (proteins – best); **epitope** – antigenic determinant; **hapten** – low-molecular substance, antigenic only with carrier (drugs – penicillin); **adjuvant** – enhances immunogenicity (**aluminium salts – alum** in DPT, TT, Hep B; **shake test** for freeze damage).
- **Antibody (immunoglobulin)** – **Y-shaped glycoprotein; 2 heavy + 2 light chains; Fab (antigen-binding) & Fc (effector)**; **IgG (75 %, crosses placenta, longest half-life ~21–23 days)**, **IgM (pentamer, largest, first in primary response, best complement activator, ABO isohaemagglutinins)**, **IgA (secretory – dimer, colostrum, mucosa)**, **IgE (allergy, parasites – mast cells)**, IgD (B-cell receptor); **made by plasma cells**.
- **Complement** – ~30 serum proteins; **classical (antigen-antibody), alternative, lectin pathways**; functions – lysis (MAC – C5b-9), opsonisation (C3b), chemotaxis (C5a), inflammation (C3a, C5a – anaphylatoxins).
- **MHC (HLA)** – class I (all nucleated cells – CD8), class II (APCs – CD4); transplant matching.
- **Hypersensitivity (Gell & Coombs)** – **Type I – immediate/ anaphylactic (IgE)**; **Type II – cytotoxic (IgG/ IgM against cell antigens – blood transfusion reaction, Rh HDN, autoimmune haemolytic anaemia, Goodpasture, Graves', myasthenia)**; **Type III – immune complex (serum sickness, Arthus, SLE, PSGN, RA)**; **Type IV – delayed, T-cell mediated (Mantoux, contact dermatitis, graft rejection, granuloma – TB)**.
- **Immunodeficiency** – primary (DiGeorge – thymic aplasia; Bruton's agammaglobulinaemia; SCID) & secondary (**HIV, malnutrition (commonest worldwide), steroids, chemotherapy, diabetes, splenectomy**).

8.2 Antigen–Antibody Reactions & Lab Tests

Reaction	Principle	Examples
Agglutination	**Particulate antigen + antibody → clumping**	**Blood grouping & cross-matching, Widal (typhoid), Weil-Felix (rickettsia), latex agglutination (RA factor, CRP, ASO), Coombs test (antiglobulin), TPHA (haemagglutination), Paul-Bunnell**, MAT (leptospira)
Precipitation	**Soluble antigen + antibody → precipitate**	**VDRL & RPR (flocculation – syphilis), Elek's gel test (diphtheria toxin)**, Ouchterlony double diffusion, radial immunodiffusion
Complement fixation	Complement consumed	Wassermann (old – syphilis)
Neutralisation	Antibody neutralises toxin/ virus	**Schick test (diphtheria immunity)**, ASO (antistreptolysin O)
ELISA (enzyme-linked immunosorbent assay)	Enzyme-labelled antibody → colour change	**HIV (screening), HBsAg, dengue NS1/ IgM, hepatitis, many**; highly sensitive
Immunofluorescence (IFA)	Fluorescent-labelled antibody	**Rabies (corneal smear/ brain), syphilis (FTA-ABS)**, autoantibodies (ANA)
Rapid tests – immunochromatography (lateral flow)/ card tests	Coloured lines	**Malaria RDT, HIV rapid, pregnancy (hCG) test, dengue, filaria ICT, COVID antigen**, rK39 (kala-azar), HBsAg rapid
Western blot	Antibodies to specific separated proteins	HIV (old confirmatory), Lyme
Molecular – PCR, RT-PCR, CBNAAT/ TrueNat, LAMP	Nucleic acid amplification – **most sensitive & specific**	**TB (GeneXpert), COVID-19 (RT-PCR – gold standard), HIV viral load/ EID, HBV/ HCV viral load**

- **Sensitivity** – ability to detect true positives (few false negatives) – **screening tests should be highly sensitive (ELISA)**; **specificity** – ability to detect true negatives (few false positives) – **confirmatory tests highly specific (Western blot, TPHA)**.

Important serological tests (one-liners)

Test	Disease
Widal	**Typhoid (O & H antibodies – rising titre 4-fold in paired sera)**
Weil-Felix	**Rickettsial infections** (heterophile – Proteus antigens)
VDRL / RPR	**Syphilis (screening, non-specific)**; **TPHA/ FTA-ABS – specific**
ASO titre	**Streptococcal infection → rheumatic fever, AGN** (> 200 IU/ml)
Paul-Bunnell / Monospot	**Infectious mononucleosis (EBV)**
Mantoux (TST)	**TB infection (type IV)**
Casoni	Hydatid (obsolete)

Test	Disease
Frei test	LGV (obsolete)
Schick test	**Susceptibility to diphtheria**
Dick test	Susceptibility to scarlet fever
Lepromin test	**Prognosis in leprosy**
Tzanck smear	**Herpes – multinucleated giant cells**
Coombs (direct)	**Haemolytic disease of newborn, AIHA**; indirect – antenatal Rh antibodies
CRP	Acute inflammation/ infection (sepsis marker – also procalcitonin)
RA factor	Rheumatoid arthritis
Nitrazine	PROM (amniotic fluid – blue) – not serology
Gravindex / UPT	Pregnancy (hCG)

8.3 Vaccines

Type	Examples	Points
Live attenuated	**BCG, OPV (bOPV), MR/ MMR, measles, rotavirus (Rotavac), varicella, yellow fever (17D), JE (SA 14-14-2), typhoid oral (Ty21a), influenza (intranasal)**	**Single dose often gives lasting immunity, mimic natural infection; contraindicated in pregnancy & severe immunodeficiency** (except specific); **most heat-sensitive (OPV – most heat-sensitive vaccine); 2 live injectable vaccines – give same day or ≥ 4 weeks apart**
Killed/ inactivated	**IPV (Salk), whole-cell pertussis, rabies (cell-culture), hepatitis A (inactivated), JE (inactivated – JENVAC), influenza (injectable), cholera (oral killed), Covaxin**	**Multiple doses + boosters needed**; safe in immunocompromised
Toxoid	**Tetanus toxoid (TT/ Td), diphtheria toxoid**	**Formalin-detoxified exotoxin; freeze-sensitive (never freeze – shake test)**
Subunit/ recombinant/ conjugate/ polysaccharide	**Hepatitis B (recombinant HBsAg – yeast), HPV, Hib (conjugate), PCV (pneumococcal conjugate), TCV (typhoid conjugate),** meningococcal, Vi polysaccharide	**Polysaccharide vaccines poor in < 2 yrs → conjugated to protein carrier**
Newer	**mRNA (COVID – Pfizer, Moderna; India – Gemcovac), viral vector (Covishield), DNA (ZyCoV-D)**	–

Universal Immunization Programme (UIP) – National Immunization Schedule (India)

Age	Vaccines
Birth	**BCG, OPV-0 (within 15 days), Hepatitis B birth dose (within 24 hrs)**
6 weeks	**OPV-1, Pentavalent-1 (DPT + HepB + Hib), Rotavirus-1, fIPV-1, PCV-1**
10 weeks	**OPV-2, Pentavalent-2, Rotavirus-2**

Age	Vaccines
14 weeks	**OPV-3, Pentavalent-3, Rotavirus-3, fIPV-2, PCV-2**
9–12 months	**MR-1, JE-1 (endemic), PCV booster, fIPV-3 (9 months – added 2023), Vitamin A 1st dose (1 lakh IU)**
16–24 months	**MR-2, JE-2, DPT booster-1, OPV booster, Vitamin A 2nd dose (2 lakh IU) – then every 6 months till 5 yrs (total 9 doses)**
5–6 years	**DPT booster-2**
10 years & 16 years	**Td**
Pregnant women	**Td-1 (early), Td-2 (4 weeks later), or Td booster** (if vaccinated within last 3 yrs)

- **Routes & sites** – **BCG – ID, left upper arm (0.1 ml; 0.05 ml < 1 month)**; **OPV – oral 2 drops**; **Pentavalent, PCV, Hep B – IM anterolateral thigh (0.5 ml)**; **fIPV – ID 0.1 ml right upper arm**; **Rotavac – oral 5 drops**; **MR – SC right upper arm (0.5 ml)**; **JE live – SC left upper arm**; **DPT booster – IM thigh**; **Td – IM upper arm**; **Vitamin A – oral (1 ml = 1 lakh IU)**.
- **Mission Indradhanush (2014)**, **Intensified MI (IMI)**; **U-WIN** (digital registry); **National Immunization Day (Pulse Polio)**; **UIP started 1985**; **EPI 1978**.
- **Missed doses** – don't restart series, continue; **Pentavalent up to 1 yr, DPT booster up to 7 yrs**; **BCG up to 1 yr**; **OPV up to 5 yrs**.

Cold chain

- **Cold chain** – system of storing & transporting vaccines at recommended temperature from manufacturer to beneficiary.
- **Equipment** – **walk-in coolers/ freezers (state/ regional)**, **ILR (ice-lined refrigerator – PHC; +2 to +8 °C; vaccines stored in baskets)**, **deep freezer (−15 to −25 °C – ice packs; OPV at state/ district level)**, **cold box**, **vaccine carrier (with 4 conditioned ice packs; keeps cold ~12 hrs+)**, **conditioned ice packs** (kept outside till "sweating"/ water sound – prevents freezing of vaccines).
- **Storage temperature** – **all UIP vaccines at +2 to +8 °C at PHC/ session level**; **OPV – −15 to −25 °C at state/ district (can be +2 to +8 °C at PHC for short)**.
- **Most heat-sensitive – OPV (then measles/ MR, BCG)**; **freeze-sensitive (never freeze) – Hep B, Pentavalent, DPT, TT/ Td, PCV, IPV, Rotavirus** ("**T-series & HepB**") – store in **ILR basket, not near bottom/ walls**; in ILR – **freeze-sensitive vaccines on top (in basket), heat-sensitive (OPV, BCG, MR) at bottom of basket**.
- **VVM (vaccine vial monitor)** – heat-sensitive label (square inside circle); **use if inner square lighter than outer circle**; **discard if square matches or darker than circle (stage 3/ 4)**.
- **Shake test** – detects **freeze damage** of adsorbed vaccines (DPT, TT, Hep B) – frozen vaccine sediments faster (flakes).
- **Open vial policy** – opened vials of **OPV, Pentavalent, DPT, TT/ Td, Hep B, IPV, PCV, Rotavirus** can be used in subsequent sessions up to **4 weeks** (if VVM OK, not contaminated, cold chain maintained, expiry not passed); **reconstituted BCG, MR, JE – discard after 4 hrs or end of session** (whichever earlier); **write date & time of opening**.
- **Diluent** – same manufacturer, cooled (+2 to +8 °C) before reconstitution; **never use water/ other diluent**.
- **Reverse cold chain** – transporting **stool samples (AFP)** in cold boxes to lab.

- **eVIN** – electronic vaccine intelligence network (real-time stock & temperature).

AEFI (Adverse Events Following Immunisation)

- **Types** – **vaccine product-related, vaccine quality defect, immunisation error-related (programmatic – commonest preventable: wrong diluent, contaminated vial, wrong site/ route, reuse of syringe)**, immunisation anxiety-related, coincidental.
- **Minor** – fever, local pain/ swelling, irritability; **BCG – papule → ulcer → scar (normal, 2–6 weeks, heals 6–12 weeks; don't apply anything)**; **BCG adenitis**.
- **Serious** – **anaphylaxis (within minutes – observe 30 min after vaccination; adrenaline 1:1000 IM 0.01 ml/kg in AEFI kit)**, **toxic shock syndrome (contaminated reconstituted vial – MR; within hours)**, encephalopathy, **VAPP (OPV)**, death, hospitalisation, cluster → **report & investigate within 24 hrs (district AEFI committee)**.
- **AD (auto-disable) syringes (0.5 ml & 0.1 ml)** – prevent reuse; **hub cutter** for disposal.
- **Contraindications** – **anaphylaxis to previous dose**; **live vaccines – pregnancy, severe immunosuppression**; **DPT – encephalopathy within 7 days of previous dose (give DT)**; **mild illness/ fever, malnutrition, diarrhoea are NOT contraindications**.

8.4 Immunoglobulins & Antisera (Passive Immunisation)

Product	Use
HTIG (human tetanus Ig) 250–500 IU (prophylaxis), 3000–6000 IU (Rx) / ATS (equine 1500 IU – test dose)	Tetanus prone wounds in non-immunised / treatment
RIG – HRIG 20 IU/kg, ERIG 40 IU/kg; rabies monoclonal antibodies	Category III animal bites
HBIG	Newborn of HBsAg+ mother, needle-stick (unvaccinated)
VZIG	Varicella exposure in immunocompromised, pregnant, newborn
Anti-D (Rho-D) 300 µg	Rh-negative mother within 72 hrs of delivery/ abortion, 28 weeks
ADS (anti-diphtheritic serum)	Diphtheria treatment
ASV (polyvalent anti-snake venom – Big 4: cobra, common krait, Russell's viper, saw-scaled viper)	Snakebite with envenomation
IVIG	Kawasaki disease, ITP, GBS, immunodeficiency
Anti-thymocyte globulin	Transplant rejection

- **Serum sickness** (type III) 7–14 days after animal serum; **anaphylaxis** – keep adrenaline ready.

Yaad karne ki trick

- Live vaccines – **"BOMR-JRY"** – **B**CG, **O**PV, **M**easles/ MR, **R**otavirus, **J**E, **Y**ellow fever (+ varicella).
- Freeze-sensitive – **"Hepa-Penta-DPT-TT never in the freezer"**.
- VVM – **"Square lighter than circle = use it; square same/ darker = discard"**.
- Open vial – **"OPV-Penta-TT 4 weeks; BCG-MR-JE 4 hours"**.

PYQ style

- First line of defence – **skin & mucous membranes**.
- Vaccination gives – **artificial active immunity**; immunoglobulin – **artificial passive**.
- Breast milk antibodies – **natural passive immunity**.
- Herd immunity needed for measles – **~95 %**.
- Booster dose principle – **secondary (anamnestic) response**.
- Adjuvant in DPT/ TT – **aluminium salt**.
- Agglutination test for typhoid – **Widal**.
- VDRL – **flocculation (precipitation)**.
- Most sensitive & specific diagnostic method – **PCR**.
- Screening test should be highly – **sensitive**.
- Rickettsial test – **Weil-Felix**; infectious mononucleosis – **Paul-Bunnell**.
- Diphtheria susceptibility – **Schick test**.
- BCG route – **intradermal**; MR – **subcutaneous**; Pentavalent – **IM thigh**.
- Hep B birth dose – **within 24 hrs**.
- Pentavalent contains – **DPT + Hep B + Hib**.
- Vitamin A first dose – **9 months (1 lakh IU)**; total doses – **9**.
- Most heat-sensitive vaccine – **OPV**.
- Freeze-sensitive vaccines – **Hep B, Pentavalent, DPT, TT**.
- Test for freeze damage – **shake test**.
- Discard vaccine when VVM – **square matches/ darker than circle**.
- Reconstituted measles/ MR vaccine discard – **after 4 hrs**.
- ILR temperature – **+2 to +8 °C**; deep freezer – **−15 to −25 °C**.
- Observe after vaccination – **30 minutes**.
- Commonest preventable AEFI – **programme (immunisation) error**.
- Toxic shock syndrome after measles vaccine – **contaminated reconstituted vial**.
- UIP started – **1985**; Mission Indradhanush – **2014**.
- Anti-D dose – **300 µg within 72 hrs**.

9 Specimen Collection, Transport & Laboratory Diagnosis

9.1 General Principles

- **Collect before starting antibiotics/ antimicrobials** (or before the next dose – trough).
- **Collect from the actual site of infection**, avoiding contamination with normal flora.
- **Adequate quantity**, at the **right time** (e.g. **early morning sputum/ urine**, **blood culture during/ before fever spike**, **stool in acute phase**).
- **Sterile, leak-proof, wide-mouthed, screw-capped containers**; **aseptic technique**; **standard precautions (gloves)**.
- **Correct labelling** – **patient's name, age/ sex, IPD/ UHID number, ward, specimen type & site, date & time of collection**, collector's name; **label the container (not the lid)**; **fill requisition form** – clinical details, provisional diagnosis, **antibiotics already given**, investigation required; **biohazard label** for high-risk (HIV/ HBV).
- **Transport promptly – ideally within 1–2 hrs**; if delay – **refrigerate at 4 °C (urine, sputum, stool – most) EXCEPT**: **CSF (meningococcus – keep at 37 °C/ room temp, never refrigerate)**, **blood culture bottles (room temp/ incubator)**, **specimens for gonococcus, anaerobes**, genital specimens.
- **Transport media** for swabs – **Stuart's/ Amies**; **Cary-Blair/ VR medium (cholera)**; **viral transport medium (VTM)** for viral swabs (COVID, influenza); **triple packaging** for infectious substances (primary leak-proof → secondary leak-proof with absorbent → outer rigid package).
- **Rejection criteria** – unlabelled/ mislabelled specimens, leaking/ broken container, insufficient quantity, wrong container/ preservative, **delayed transport**, **24-hr urine incomplete**, saliva instead of sputum, dry swabs, formalin-fixed tissue for culture.

9.2 Blood Culture

- **Indications** – fever of unknown origin, **sepsis/ septic shock**, endocarditis, typhoid (1st week), neonatal sepsis, pneumonia, meningitis.
- **Timing** – **before antibiotics**, **ideally at onset of fever/ chills**; **2 sets (aerobic + anaerobic bottles) from 2 different sites** (to distinguish contaminants); **endocarditis – 3 sets** over hrs.
- **Volume** – **adults 8–10 ml per bottle (20 ml/ set)**; **children 1–5 ml (paediatric bottles)**; neonates 0.5–1 ml; **blood : broth ratio 1 : 5 to 1 : 10**.
- **Technique** – hand hygiene, gloves; **disinfect skin with 70 % alcohol then chlorhexidine-alcohol (or povidone-iodine – allow to dry 1–2 min)**; **don't palpate site after disinfection**; disinfect bottle tops with alcohol; **venepuncture, not from existing IV line** (except catheter-related BSI with paired peripheral sample); **inoculate aerobic bottle first (if using syringe – anaerobic first if butterfly to avoid air)**; label; **don't refrigerate** – send immediately/ incubate at 37 °C.
- **Automated systems (BACTEC, BacT/ALERT)** – continuous monitoring; most positives in 24–48 hrs; final negative at 5 days.
- **Common contaminants** – **CoNS (S. epidermidis), diphtheroids, Bacillus, Propionibacterium** (skin flora).

9.3 Urine

Specimen	Method
Clean-catch midstream urine (MSU) – commonest	**Early morning sample (concentrated) preferred**; **wash genitals with soap & water (no antiseptic)** – female: separate labia, clean front to back; male: retract foreskin; **discard first part, collect middle portion (~10–20 ml) in sterile container**, don't touch inside of container
Catheter specimen (CSU)	**Never from drainage bag**; **clamp tubing below sampling port for 15–30 min**, **disinfect sampling port with alcohol**, **aspirate with sterile needle/ syringe from port** (or proximal catheter wall)
Suprapubic aspiration	**Gold standard (sterile) – infants**; any growth significant
Infants	Sterile adhesive urine bag (screening only), clean-catch, catheter
24-hour urine	**Discard first morning void, note time, collect all subsequent urine for 24 hrs including next morning's first void at same time**; preservative as per test; refrigerate; **if a sample is missed – restart**
For AFB (genitourinary TB)	**Early morning urine × 3–5 consecutive days** (entire first void)

- **Transport** within **1–2 hrs** or **refrigerate at 4 °C (up to 24 hrs)** – bacteria multiply at room temp → false +ve.
- **Kass criteria – significant bacteriuria ≥ 10^5 CFU/ml (100,000/ml)** in MSU; **10^3–10^5 – doubtful (repeat)**; **< 10^3 – contamination/ insignificant**; any count in SPA significant.
- **Urine routine** – **pus cells > 5/HPF – pyuria**; **nitrite test (dipstick) – Gram-negative bacteria (E. coli)**; leukocyte esterase.

9.4 Stool

- **Fresh stool (~2–5 g or walnut size; 5–10 ml liquid)** in **clean, dry, wide-mouthed, screw-capped container**; **not mixed with urine, water or toilet disinfectant**; **from portion with blood/ mucus**.
- **For trophozoites (amoeba, Giardia) – examine within 30 min–1 hr (warm)**; **for ova & cysts – 3 samples on alternate/ consecutive days**.
- **Rectal swab** – when stool not available (cholera, shigella in infants).
- **Transport** – **Cary-Blair (cholera, Salmonella, Shigella)**, **VR medium/ alkaline peptone water (cholera)**, **buffered glycerol saline (Shigella)**; **AFP surveillance stool – 2 samples (≥ 8 g each) 24 hrs apart within 14 days of paralysis, reverse cold chain (2–8 °C)**.
- **Occult blood** – avoid red meat, iron, vitamin C, NSAIDs before test (guaiac).
- **Pinworm – cellophane tape (NIH swab), early morning before bathing/ defecation**.

9.5 Respiratory Specimens

- **Sputum** – **early morning, deep cough specimen (not saliva/ nasal secretion)**; **rinse mouth with plain water (no antiseptic/ toothpaste)**; take deep breaths, cough forcefully; **2–5 ml (purulent/ mucopurulent)** in **sterile wide-mouthed container**; **collect outdoors/ well-ventilated area** (TB); **TB – 2 samples (spot + early morning) for microscopy; 1 for CBNAAT/ TrueNat (NTEP)**; **induced sputum (3 % hypertonic saline nebulisation)** if unable; **gastric aspirate/ lavage – children (early**

morning, empty stomach); **quality – < 10 squamous epithelial cells & > 25 pus cells/ LPF (Bartlett's/ Murray-Washington)**.

- **Throat swab** – **tongue depressed, swab both tonsils & posterior pharyngeal wall (avoid touching tongue, cheeks, teeth)** – **diphtheria (both cotton & from edge of membrane)**, **streptococcal pharyngitis**.
- **Nasopharyngeal swab** – **flexible flocked swab through nostril parallel to palate to nasopharynx (depth ≈ nostril to ear tragus), rotate & hold few seconds** – **influenza, COVID-19 RT-PCR, pertussis (pernasal swab)**, RSV; place in **VTM, 2–8 °C**.
- **Nasal swab** – MRSA carrier screening (anterior nares).
- **BAL/ endotracheal aspirate** – ventilated patients (VAP).

9.6 Other Specimens

Specimen	Key points
Pus & wound swab	**Aspirated pus (syringe) better than swab** (more organisms, anaerobes); **clean wound surface with sterile saline first (remove surface debris & exudate)**, **swab from base/ advancing edge using Levine technique (rotate over 1 cm^2 with pressure)**; 2 swabs (Gram stain + culture); anaerobes – **transport in syringe/ anaerobic medium without air**; deep tissue biopsy best
CSF	**Lumbar puncture by doctor (L3–L4/ L4–L5), strict asepsis**; **3 sterile tubes – 1st biochemistry (glucose, protein), 2nd microbiology (Gram stain, culture – most important), 3rd cytology/ cell count**; **~1 ml each (3–5 ml)**; **transport immediately, never refrigerate (keep at 37 °C – meningococcus & H. influenzae are fragile)**; India ink (Cryptococcus), CBNAAT (TB)
High vaginal swab (HVS)/ endocervical swab	**Speculum (no lubricant except water)**; HVS – Candida, Trichomonas (wet mount – examine immediately/ warm), bacterial vaginosis (**clue cells, whiff test – fishy odour with KOH, Amsel criteria**); **endocervical – gonococcus (Stuart's/ Amies, room temp, not refrigerated), Chlamydia (NAAT)**
Urethral swab	Gonococcus – **Gram stain (intracellular Gram –ve diplococci)**, culture
Eye swab/ conjunctival	Before instilling drops; ophthalmia neonatorum
Ear swab	Otitis externa/ media discharge
Skin scrapings, nail, hair	**Clean with 70 % alcohol, scrape advancing edge with scalpel, collect in black paper/ envelope (not closed container – moisture)** – fungi (KOH)
Blood for serology	**3–5 ml in plain tube (red top – no anticoagulant) → serum**; paired sera 10–14 days apart (rising titre)
Blood smear for malaria	**Finger prick**; thick & thin smears **on same slide/ separate**, **air-dry, label**; collect at fever/ before antimalarial; **thick smear – not fixed (dehaemoglobinised); thin smear – fixed with methanol**
Microfilaria	**Night blood (10 PM–2 AM)** – thick smear
Tissue/ biopsy	**Culture – in sterile saline (not formalin)**; histopathology – **10 % formalin (10 × volume)**

Specimen	Key points
Blood – anticoagulants (vacutainer colour)	**Purple/ lavender – EDTA (CBC, blood group, HbA1c)**; **light blue – sodium citrate 3.2 % (PT/ INR, aPTT – 1:9 ratio, fill exactly)**; **red – plain/ clot activator (serum – serology, biochemistry)**; **yellow/ gold – SST gel (serum chemistry)**; **green – heparin (electrolytes, ABG – heparinised syringe)**; **grey – sodium fluoride + potassium oxalate (blood glucose – inhibits glycolysis)**; **ESR – black (citrate 1:4)**
Order of draw (CLSI)	**Blood culture → light blue (citrate) → red/ gold (serum) → green (heparin) → lavender (EDTA) → grey (fluoride)**

9.7 Laboratory Safety & Biosafety

- **Biosafety levels (BSL)** – **BSL-1** (non-pathogenic – B. subtilis, E. coli K-12; teaching labs); **BSL-2** (moderate risk – **most clinical labs – S. aureus, HBV, HIV (specimens), Salmonella**; class II BSC for aerosols); **BSL-3** (serious airborne – **M. tuberculosis cultures, SARS-CoV-2 culture, Brucella, Yersinia pestis**; negative pressure, BSC class II/ III); **BSL-4** (life-threatening, no treatment – **Ebola, Marburg, Nipah, smallpox, Lassa**; positive-pressure suits; **NIV Pune – India's BSL-4 lab**).
- **Biological safety cabinets** – **class I** (protects user & environment), **class II (protects user, product & environment – most common in clinical labs, TB lab)**, **class III** (glove box – BSL-4).
- **Lab rules** – no eating/ drinking/ mouth pipetting; lab coat & gloves; hand wash; disinfect bench (start & end of day); report spills & accidents; **sharps safety**; vaccinated against HBV.
- **ICMR** – apex body for biomedical research; **NIV Pune (virology), NICD → NCDC Delhi (surveillance), NIRT Chennai (TB), NIMR Delhi (malaria)**, **Central Research Institute Kasauli (vaccines)**, **BCG Lab Guindy (Chennai)**, **Pasteur Institute Coonoor (rabies)**, **Haffkine Institute Mumbai (plague vaccine – W. M. Haffkine)**; **IDSP (Integrated Disease Surveillance Programme, 2004) – S, P, L forms (Syndromic, Presumptive, Laboratory)**; **IHIP portal**.

9.8 Nurse's Role in Microbiology

1. **Explain procedure** to patient, obtain cooperation/ consent.
2. **Collect correct specimen** at correct time, **before antibiotics**, using correct container & technique.
3. **Aseptic technique & standard precautions**; avoid contamination.
4. **Label correctly** & complete requisition form; **mention antibiotics**.
5. **Transport immediately** under appropriate conditions; **maintain chain of custody**.
6. **Record** collection in nurses' notes; **follow up & communicate reports** (critical values) to doctor.
7. Infection control – isolation, hand hygiene, waste disposal, **report notifiable diseases**.
8. **Health education** – hand washing, safe water & food, immunisation, completion of antibiotics.

Yaad karne ki trick

- CSF – **"Never chill the brain"** – never refrigerate CSF.
- CSF tubes – **"1 Biochem, 2 Bugs (micro), 3 Cells"**.
- Significant bacteriuria – **"Lakh = Lakshya" – ≥ 1 lakh (10^5) CFU/ml**.
- Vacutainer – **"Lavender = Lab CBC (EDTA), Blue = Bleeding (coagulation), Grey = Glucose, Green = heparin (gas/ electrolytes), Red = seRum"**.

PYQ style

- Specimen collection – **before antibiotics**.
- Urine culture – **early morning midstream clean-catch**.
- Catheter urine – **from sampling port with sterile syringe, never from bag**.
- Significant bacteriuria – **≥ 10^5 CFU/ml**.
- Urine transport delay – **refrigerate 4 °C**.
- Gold standard urine sample in infants – **suprapubic aspiration**.
- Sputum for TB – **early morning, deep cough, rinse mouth with water**.
- Children TB specimen – **gastric aspirate**.
- Blood culture – **2 sets, 2 sites, 8–10 ml per bottle (adult)**.
- Blood culture skin prep – **chlorhexidine-alcohol/ povidone-iodine (let dry)**.
- Commonest blood culture contaminant – **CoNS**.
- CSF – **never refrigerate; send immediately**.
- Stool for amoebic trophozoites – **examine within 30 min**.
- Cholera stool transport – **Cary-Blair/ VR medium**.
- Pinworm – **cellophane tape test in early morning**.
- Microfilaria blood – **night (10 PM–2 AM)**.
- Nasopharyngeal swab transport – **VTM at 2–8 °C**.
- Pus culture – **aspirate better than swab**.
- Tissue for culture – **sterile saline, not formalin**.
- Blood glucose tube – **grey (fluoride)**; CBC – **lavender (EDTA)**; PT/INR – **light blue (citrate)**.
- First in order of draw – **blood culture**.
- TB culture lab – **BSL-3**; Ebola, Nipah – **BSL-4 (NIV Pune)**.
- Most common BSC – **class II**.
- IDSP launched – **2004**.

10 Quick Revision – Master Tables & 150 One-Liners

10.1 Disease → Causative Organism

Disease	Organism
Boils, abscess, osteomyelitis	**Staphylococcus aureus**
Rheumatic fever, scarlet fever	**Streptococcus pyogenes (Group A)**
Lobar pneumonia	**Streptococcus pneumoniae**
Gonorrhoea / meningococcal meningitis	**Neisseria gonorrhoeae / N. meningitidis**
Diphtheria	**Corynebacterium diphtheriae**
Tetanus / gas gangrene / botulism	**Clostridium tetani / C. perfringens / C. botulinum**
Anthrax	**Bacillus anthracis**
Typhoid	**Salmonella Typhi**
Bacillary dysentery	**Shigella**
Cholera	**Vibrio cholerae**
UTI	**E. coli**
Whooping cough	**Bordetella pertussis**
Peptic ulcer	**Helicobacter pylori**
Plague	**Yersinia pestis**
TB / leprosy	**Mycobacterium tuberculosis / M. leprae**
Syphilis	**Treponema pallidum**
Leptospirosis	**Leptospira interrogans**
Scrub typhus	**Orientia tsutsugamushi**
Trachoma	**Chlamydia trachomatis**
Atypical pneumonia	**Mycoplasma pneumoniae**
Chancroid	**Haemophilus ducreyi**
Infectious mononucleosis	**EBV**
Cervical cancer	**HPV 16, 18**
AIDS	**HIV**
Rabies	**Rabies virus (Lyssavirus)**
Kala-azar	**Leishmania donovani**
Amoebic dysentery	**Entamoeba histolytica**
Ringworm	**Dermatophytes (Trichophyton, Microsporum, Epidermophyton)**
Thrush	**Candida albicans**

Disease	Organism
Elephantiasis	**Wuchereria bancrofti**

10.2 Incubation Periods

Disease	Incubation
Staph food poisoning	**1–6 hrs**
Cholera	**Few hrs – 5 days (1–2 days)**
Influenza	**1–4 days**
Diphtheria	**2–6 days**
Dengue	**4–7 days (3–14)**
Tetanus	**6–10 days (3–21)**
Typhoid	**10–14 days (3–21)**
Measles	**10 days (to fever), 14 days (to rash)**
Malaria (vivax)	**~14 days**
Chickenpox	**14–16 days**
Mumps	**16–18 days**
Rubella	**14–21 days (~18)**
Hepatitis A	**15–45 days (~30)**
Hepatitis B	**30–180 days (~75)**
Rabies	**1–3 months (variable)**
Leprosy	**3–5 years** (longest)

10.3 Important Days

Date	Day
30 January	**Anti-Leprosy Day** (Gandhi's death anniversary)
24 March	**World TB Day**
7 April	World Health Day
25 April	**World Malaria Day**
5 May	**World Hand Hygiene Day**
16 May	**National Dengue Day**
14 June	World Blood Donor Day
28 July	**World Hepatitis Day** (Baruch Blumberg – HBsAg)
28 September	**World Rabies Day**
15 October	Global Handwashing Day
18–24 November	**World AMR Awareness Week**

Date	Day
1 December	World AIDS Day
24 October	World Polio Day

10.4 150 Rapid-Fire One-Liners

150 one-liners

1. Father of microbiology – **Leeuwenhoek**.
2. Father of modern microbiology – **Pasteur**.
3. Father of bacteriology – **Koch**.
4. Father of antiseptic surgery – **Lister**.
5. Father of immunology – **Jenner**.
6. Father of chemotherapy – **Ehrlich**.
7. Germ theory – **Pasteur**.
8. Penicillin – **Fleming (1928)**.
9. Gram stain – **Hans Christian Gram (1884)**.
10. TB bacillus – **Koch (1882)**.
11. Leprosy bacillus – **Hansen (1873)**.
12. Viruses discovered – **Ivanovsky**.
13. Prokaryotic ribosome – **70S**.
14. No cell wall – **Mycoplasma**.
15. Infectious protein – **prion**.
16. Bacteria size – **µm**; virus – **nm**.
17. Virus seen by – **electron microscope**.
18. Syphilis spirochaete seen by – **dark-ground microscopy**.
19. Oil immersion – **100× objective**.
20. Skin flora – **S. epidermidis**.
21. Gut flora – **Bacteroides**.
22. Vaginal flora – **Lactobacilli**.
23. Grape clusters – **Staphylococci**.
24. Chains – **Streptococci**.
25. Comma-shaped – **Vibrio**.
26. Chinese letters – **C. diphtheriae**.
27. Bacterial cell wall – **peptidoglycan**.
28. Endotoxin – **LPS of Gram-negative**.
29. Capsule – **antiphagocytic**.
30. Motility organ – **flagella**.
31. Conjugation – **sex pilus**.
32. Resistance genes – **plasmids**.
33. Spore formers – **Bacillus, Clostridium**.
34. Drumstick spore – **C. tetani**.
35. Gram +ve colour – **purple**.
36. Gram −ve colour – **pink**.
37. Gram mordant – **iodine**.
38. AFB stain – **Ziehl-Neelsen**.
39. Albert stain – **diphtheria**.
40. India ink – **Cryptococcus**.

41. KOH mount – **fungi**.
42. Giemsa – **malaria**.
43. Antibiotic-sensitive phase – **log phase**.
44. E. coli generation time – **20 min**.
45. TB culture – **LJ medium**.
46. Cholera medium – **TCBS**.
47. Fungi medium – **SDA**.
48. Lactose fermenter (pink) – **E. coli on MacConkey**.
49. Chocolate agar – **Haemophilus, Neisseria**.
50. Anaerobic medium – **Robertson's cooked meat**.
51. AST method – **Kirby-Bauer (Mueller-Hinton)**.
52. Most potent toxin – **botulinum**.
53. Toxoid – **tetanus, diphtheria**.
54. MRSA gene – **mecA**.
55. Sterilisation – **kills spores**.
56. Most resistant – **prions**.
57. Autoclave – **121 °C, 15 psi, 15 min**.
58. Hot air oven – **160 °C, 2 hrs**.
59. Autoclave biological indicator – **G. stearothermophilus**.
60. Pasteurisation holder – **63 °C × 30 min**.
61. Flash pasteurisation – **72 °C × 15–20 s**.
62. Tyndallisation – **3 successive days**.
63. Disposable syringes – **gamma radiation**.
64. Endoscopes – **2 % glutaraldehyde**.
65. Heat-sensitive plastics – **ETO**.
66. OT fumigation – **formalin + $KMnO_4$**.
67. Blood spill – **1 % hypochlorite**.
68. Best alcohol – **70 %**.
69. Phenol coefficient – **Rideal-Walker**.
70. Critical items – **sterilisation**.
71. Semi-critical – **HLD**.
72. Non-critical – **LLD**.
73. HAI after – **48 hrs**.
74. Commonest HAI – **CAUTI**.
75. Most effective infection control – **hand hygiene**.
76. Hand rub time – **20–30 s**.
77. Hand wash time – **40–60 s**.
78. 5 moments – **WHO**.
79. TB isolation – **airborne, N95, negative pressure**.
80. Influenza – **droplet precautions**.
81. MRSA – **contact precautions**.
82. First PPE off – **gloves**.
83. NSI highest risk – **HBV**.

84. HIV PEP – **within 72 hrs, 28 days**.
85. Sharps – **white container**.
86. Anatomical waste – **yellow bag**.
87. Contaminated plastics – **red bag**.
88. Glassware – **blue box**.
89. BMW storage limit – **48 hrs**.
90. Coagulase-positive – **S. aureus**.
91. ASO – **streptococcal**.
92. Dental caries – **S. mutans**.
93. Optochin-sensitive – **pneumococcus**.
94. Kidney-shaped diplococci – **Neisseria**.
95. Pseudomembrane – **diphtheria**.
96. Diphtheria death – **myocarditis**.
97. Woolsorter's disease – **anthrax**.
98. Lockjaw – **tetanus**.
99. Pseudomembranous colitis – **C. difficile**.
100. Typhoid 1st week – **blood culture**.
101. Widal – **2nd week**.
102. Typhoid carrier site – **gall bladder**.
103. Rice-water stool – **cholera**.
104. Green pus – **Pseudomonas**.
105. X & V factors – **H. influenzae**.
106. Whooping cough – **B. pertussis**.
107. Urease breath test – **H. pylori**.
108. Plague vector – **rat flea**.
109. Mantoux reading – **48–72 hrs**.
110. Mantoux positive – **≥ 10 mm**.
111. BCG – **live M. bovis, intradermal**.
112. Rapid TB test – **CBNAAT**.
113. Leprosy cultured in – **armadillo**.
114. Leprosy nerve – **ulnar**.
115. Syphilis screening – **VDRL**.
116. Syphilis Rx – **benzathine penicillin**.
117. Leptospirosis – **rat urine**.
118. Scrub typhus – **mite, eschar**.
119. Trachoma – **SAFE strategy**.
120. Negri bodies – **rabies**.
121. Koplik's spots – **measles**.
122. Owl's eye – **CMV**.
123. HBV first marker – **HBsAg**.
124. HBV infectivity – **HBeAg**.
125. Hepatitis E danger – **pregnancy**.
126. Most chronic hepatitis – **C**.

127. Infant diarrhoea – **rotavirus**.
128. Polio-free India – **27 March 2014**.
129. Measles death – **pneumonia**.
130. CRS – **cataract, PDA, deafness**.
131. Pandemic flu – **antigenic shift**.
132. Rabies first aid – **soap & water 15 min**.
133. Dengue vector – **Aedes (day biter)**.
134. JE amplifier – **pigs**.
135. HIV enzyme – **reverse transcriptase**.
136. AIDS – **CD4 < 200**.
137. Fungal membrane – **ergosterol**.
138. Black fungus – **mucormycosis**.
139. Anchovy sauce pus – **amoebic liver abscess**.
140. Kala-azar vector – **sandfly**.
141. Toxoplasma host – **cat**.
142. Malaria definitive host – **mosquito**.
143. Malaria relapse – **hypnozoites**.
144. Hookworm – **anaemia**.
145. Pinworm – **Scotch tape test**.
146. Hydatid – **dog tapeworm, liver**.
147. Live vaccine – **OPV, BCG, MR**.
148. Most heat-sensitive vaccine – **OPV**.
149. Freeze damage test – **shake test**.
150. CSF specimen – **never refrigerate**.



Exam strategy – Asha Didi ki last advice

- GSSSB Staff Nurse paper me **Microbiology se ~5–8 direct questions** aate hain, aur Community Health/ MSN/ Fundamentals ke infection-control questions bhi isi se jude hote hain.
- Sabse zyada pooche jaate hain – **scientists & discoveries, sterilisation (autoclave, hot air oven, glutaraldehyde, hypochlorite), BMW colour coding, hand hygiene & isolation precautions, stains & culture media, disease–organism, vaccines & cold chain, specimen collection**.
- **Numbers yaad rakho** – 121 °C/15 psi/15 min, 160 °C/2 hrs, 1 % hypochlorite, 70 % alcohol, 48 hrs BMW, 10^5 CFU/ml, Mantoux 10 mm.
- MCQ Bank PDF ke saare questions **2 baar** solve karo aur galat hue questions ko notebook me likho.

— Microbiology · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —